INSTITUT FÜR INFORMATIK
DER LUDWIG–MAXIMILIANS–UNIVERSITÄT MÜNCHEN

Bachelorarbeit

Theoretical Study of Stratified Photon Mapping with Kernel

Evert Andrés Buzón Badillo

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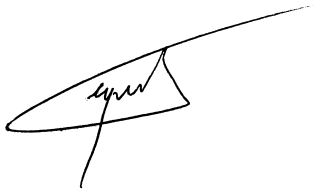

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Abstract

In this work, the over- and under-estimation biases in the irradiance estimation of Photon Mapping, firstly reported in [Gar12] by Dr. García, are studied. In particular, these biases are studied for stratified Photon Mapping with kernels, adopting some of the stratification strategies used in [Har20]. For this purpose, the existence and behavior of the biases are tested for four of the kernels (Epanechnikov, Cone, Silverman, and Gaussian) used in [Gar14] and [Gan18]. Here is mathematically and empirically demonstrated that the over- and under-estimation biases in stratified Photon Mapping can not be eliminated using kernels in general, however it is shown that a reduction of these biases is possible if kernels are used.
Also, the improvement in the standard deviation for the different kernels is empirically studied under different stratification strategies. A stratification of the scene is simulated by stratifying the light source, aiming to generate a secondary stratification on the scene. In particular, the position and direction of emission of the photons are stratified.

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1. Introduction

This work aims to extend the work done in García et al. [Gar14] and [Har20] by combining the studies on the over- and under-estimation biases of the Photon Mapping algorithm, an algorithm that is used to simulate realistic illumination ([Jen96a]), using filtering kernels and stratified sampling, a sampling technique for improving the variance.
Section 1 gives a brief introduction to the topics of computer graphics, photon mapping, and background knowledge in mathematics and statistics. A summary of previous works on Photon Mapping will also be presented. In section 2 the effects of the over- and under-estimation biases in the stratified Photon Mapping algorithm using filtering kernels will be theoretically studied. Section 3 contains an empirical approach to the study done in section 2 and also the viability of different stratification strategies will be tested on a real scene; finally, section 4 presents conclusions and a plan for future work.

1.1. Mathematical and statistical prerequisites

To study the over- and under-estimation biases on a theoretical level it is essential to mathematically model the irradiance estimation of the Photon Mapping algorithm (Sec. 1.3), this can be achieved with the help of integrals and key statistical concepts like for example the expected value. Also, for the empirical part of this work, it is necessary to efficiently generate uniformly distributed points (photons) for the different stratification strategies, for this task curvilinear coordinates will be used.

1.1.1. Curvilinear coordinates

Probably the most common coordinate system used in mathematics is the cartesian coordinate system, however, it is sometimes useful to represent a mathematical expression or formula in other coordinate systems. In particular, for this work, it was in some cases very convenient to make use of two specific curvilinear coordinate systems, namely polar coordinates (for 2D expressions) and spherical coordinates (for 3D expressions).

Polar

The relation between polar and cartesian coordinates is the following:

$$r = \sqrt{x^2 + y^2}$$
$$\theta = \arctan(y/x)$$

$$x = r\cos\theta$$
$$y = r\sin\theta$$

Where $r \in \mathbb{R}_{\geq 0}$ and $\theta \in [0, 2\pi]$.

Spherical

The relation between spherical and cartesian coordinates is:

$\rho = \sqrt{x^2 + y^2 + z2}$
$\theta = \arctan(y/x)$
$\phi = \arccos(z/\rho)$

$x = \rho \sin\varphi \cos\theta$
$y = \rho \sin\varphi \sin\theta$
$z = \rho \cos\varphi$

Where $\rho \in \mathbb{R}_{\geq 0}$, $\theta \in [0, 2\pi]$ and $\varphi \in [0, \pi]$.

1.1.2. Integrals

Here the definitions of integrals in cartesian, polar, and spherical coordinates will be presented. In this work, the integrals in cartesian coordinates are used to calculate the expected value of the irradiance estimate, and the integrals in polar and spherical coordinates are mainly used to calculate the normalization constants of the different kernels:

Cartesian 2D

$\iint_A f(x, y)\, dx\, dy$

Cartesian 3D

$\iiint_V f(x, y, z)\, dx\, dy\, dz$

Polar

$\iint_A f(r, \theta) r\, dr\, d\theta$

Spherical

$\iiint_V f(\rho, \theta, \varphi) \rho^2 \sin\varphi\, d\rho\, d\theta\, d\varphi$

Where A stands for the integration intervals of an area and V stands for the integration intervals of a volume.

1.1.3. Expected value

The expected value can be calculated for continuous, discrete, and mixed random variables (a variable that has a continuous part and a discrete part), however, in this work, only the continuous case is of relevance. Since Photon Mapping is a probabilistic algorithm, the result of the irradiance estimate may vary each run, for this reason, the algorithm has to be executed several times and the final result of the irradiance estimate is the average result of the iterations. On a theoretical level, the average result of the iterations can be modeled

by the expected value of the irradiance estimate. The definition of the expected value for a continuous random variable is:

$$E[X] = \int f(x)x\,dx \tag{1.1}$$

Where X is the value of the random variable and $f(x)$ is the probability density function of that variable.
A very interesting property of the expected value, that is going to be very useful in this study, is:

$$E[X + Y] = E[X] + E[Y] \tag{1.2}$$

Where X and Y are two random variables.

1.2. Computer graphics

Computer graphics is the field of computer science, whose goal is to generate different types of images with the help of computing power. Some of the topics in computer graphics are rendering, geometry processing, computer animation, and illumination, being illumination of special interest for the work that is presented in the following sections of this study.
In particular, global illumination is very important for this work, since the Photon Mapping algorithm is a global illumination technique.

1.3. Photon mapping

A very hard problem in computer graphics was to efficiently simulate diffuse interreflections and caustics, examples of caustics can be found in Figures 1.1 and 1.2, but this problem was finally efficiently solved by the Photon Mapping algorithm, making scenes with caustics practical. The Photon Mapping algorithm can be divided into two parts, the photon map, and the rendering part.

In the photon map part, photons are generated from light sources and stored when they hit non-specular objects, the rendering part consists of a set of calculations to be performed to get information about the incoming flux (based on the information of incoming flux of the surrounding photons stored in the photon map) and reflected irradiance at any point in the scene (this work will be concentrated on the rendering part).
Probably the most important advantage of Photon Mapping compared to other global illumination algorithms is the efficiency of the algorithm, although more memory is needed for generating the photon map.

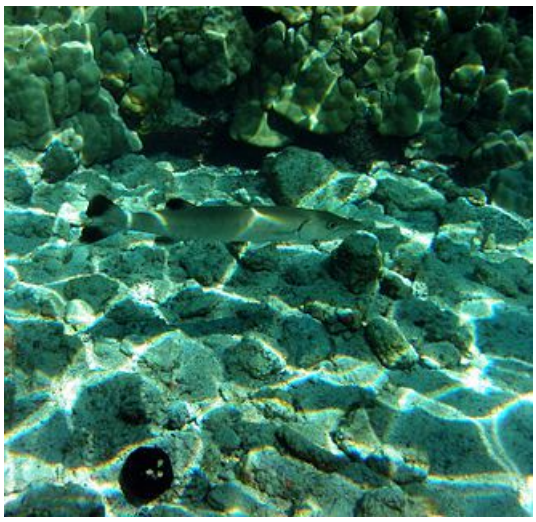


Figure 1.1.: Caustics generated by the interaction between light and the surface of water. Author: Brocken Inaglory, https://en.wikipedia.org/wiki/File:Great_Barracuda,_corals,_sea_urchin_and_Caustic_(optics)_in_Kona,_Hawaii_2009.jpg

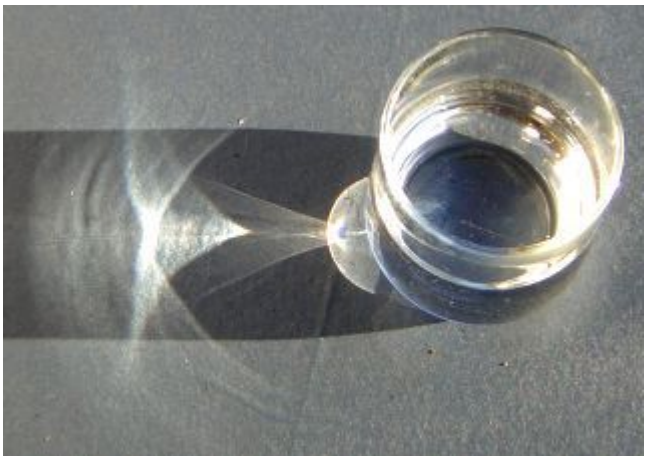


Figure 1.2.: Caustics generated by the interaction between light and a glass of water. Author: Heiner Otterstedt, https://en.wikipedia.org/wiki/File:Kaustik.jpg

1.3.1. Irrandiance estimate

In Photon Mapping there are two physical quantities of particular importance, the flux, and the irradiance. The flux ϕ can be described as the rate of flow of radiant energy of light and the irradiance I is the flux per unit area (in this work the standard symbol for the irradiance E is not used because this symbol is preserved for the expected value). In the rendering part of the Photon Mapping algorithm, as mentioned before, we want to estimate the incoming flux and the reflected irradiance at any given point in the scene, this can be done by taking into account the information of incoming flux of the k nearest photons stored in the photon map data structure, the formula for irradiance estimate $\hat{I}(r_k)$ used in Photon Mapping is:

$$\hat{I}(r_k) = \frac{k\phi}{\pi r_k^2} \tag{1.3}$$

Where r_k is the distance from the kth nearest photon impact to the point where the irradiance estimate is being calculated.
The irradiance is of particular importance for this work since the bias to be studied manifests when estimating the irradiance.

1.3.2. Kernels

In most algorithms in computer graphics kernels are used as filters to blur an image and reduce the high-frequency noise. In Photon Mapping, apart from that, kernels are used to give more weight to photons near the point of interest. This is particularly convenient when dealing with caustics [Jen02].
In this work four different kernels will be studied for the stratified case of Photon Mapping, they are presented here in their unnormalized version (the normalization process will be explained in the theoretical part of this work Sec. 2). The symbols r_i and r_k are the distances from the ith and k^{th} nearest photon impacts to the point, where the irradiance is being estimated:

Epanechnikov Kernel

The Epanechnikov kernel was first used in the context of Photon Mapping by [Wal98].

$$1 - \left(\frac{r_i}{r_k}\right)^2 \tag{1.4}$$

Cone Kernel

The Cone kernel was used by Jensen in [Jen96b].

$$1 - \frac{r_i}{sr_k} \tag{1.5}$$

Where s is the slope of the Cone kernel.

Silverman Kernel

The Silverman kernel was used by P.Shirley [Shi95].

$$\left(1 - \left(\frac{r_i}{r_k}\right)^2\right)^2 \tag{1.6}$$

Gaussian Kernel

The Gaussian kernel was used by Jensen [Jen96b].

$$1 - \frac{1 - e^{-\frac{\beta r_i^2}{2r_k^2}}}{1 - e^{-\beta}} \tag{1.7}$$

Where $\beta = 1.953$.

1.3.3. Irradiance estimate with kernel

When using kernels the irradiance estimate can be obtained as follows:

$$\hat{I}(r_1...r_k) = \sum_{i=1}^{k} K_{r_k}(r_i)\phi \tag{1.8}$$

Where $K_{r_k}(r_i)$ is a normalized filtering kernel.

1.4. Stratified sampling

Sampling in statistics is the process of selecting a specific number of individuals from a larger population. In concrete, stratified sampling is a sampling technique that divides a surface into N disjunct strata of same area, in each stratum a number x of random samples proportional to the area of the stratum is taken (in photon mapping a sample means a photon). In Photon Mapping stratified sampling is used to reduce the variance in the irradiance estimate.

1.5. Previous work on photon mapping

For the first time in [Gar12] a new type of bias in the Photon Mapping algorithm was detected, this new bias consisted of an overestimation (and in some cases underestimation) when calculating the irradiance in the algorithm. In that study Dr. García demonstrated this bias for a constant kernel, also it was shown that this bias could be eliminated for a random sampling strategy with a constant kernel taking into account for the irradiance estimation only the $(k-1)$ nearest photons and not the k nearest photons. Later in [Gar14] the same bias was evidenced for non-constant kernels and it was also shown that for random sampling the bias could be eliminated using kernels, then in [Gan18] the variance was calculated for both cases constant and non-constant kernels and in [Har20] the bias was studied for a stratified version of the Photon Mapping algorithm, however for stratified sampling was not possible to eliminate the bias, it was only possible to reduce it with the technique used in [Gar12], finally in [Sch21] the variance for the stratification strategies used in [Har20] was studied.

Photon Mapping Parameters	
P	Point where the photon map query is done
n	Number of photon impacts
k	Number of photon impacts sought in the photon map query
r_k	Distance of the kth nearest photon in a query.
ϕ	Flux
Distribution Functions	
$f_{X_{(k)}}(r_k)$	Distribution function for the kth photon impact
$f_{XY}(r_i, r_k)$	Distribution function for the ith and k^{th} photon impacts
Irradiance	
I	Irrandiance
$\hat{I}$	Irradiance estimated by Photon Mapping
$\hat{I}_i$	Estimate of the contribution of the ith nearest photon to $\hat{I}$.
PD	Power density
$\widehat{PD}$	Power density estimate
$\widehat{PD}_i$	Estimate of the contribution of the ith nearest photon to $\widehat{PD}$.
$K_{r_k}(r_i)$	Normalized filtering kernel
C_x	Normalization constant of the kernel x

Figure 1.3.: Symbols used in this thesis, some of which were adopted from [Gar14] for consistency reasons.

To study the over- and under-estimation biases the experiment done in [Gar12] will be reproduced, in this experiment a disk of radius 1 will be used as the scene for the theoretical study of both cases without filter and with kernel, the irradiance estimate will be calculated only for the point P located in the center of the disk. Following the so-called basic stratification strategy (Figure 1.5) introduced in [Har20] the disk is divided (stratified) in n strata, which is also the number of photons in the scene, each stratum has the same area in the 2D case (π/n) or volume in the 3D case (4π/3n). The radii of the circles corresponding to a specific ring that constitutes a stratum are defined by the formula t(i)=$\sqrt{i/n}$ in 2D or T(i) = $\sqrt[3]{i/n}$ in 3D, where i is the ith nearest ring to the center of the disk. The relation between the two circles that build a ring (stratum) at the position i is t(i) for the outer circle and t(i-1) for the inner circle, this ensures that the area of every stratum is the same. To also ensure that the photons emitted from the light source follow this stratification strategy, the position of emission of the photons is stratified following the basic stratification strategy. In every stratum, a single photon is emitted in a random position under a uniform distribution. Unfortunately, stratifying the position of emission alone does not guarantee that the photons will land on the scene in the same position, in which they were emitted, since the direction of emission could change their landing position. For this reason, the photons have to be emitted in a direction perpendicular to the scene, this can be achieved by a directional light source with a uniform distribution of emissive power of light (following the basic stratification strategy). In [Har20] another stratification strategy besides the simple stratification strategy was introduced, the so-called stratified sampling with order statistics (Figure 3.14), however, this strategy will only be used in the empirical part of this work. There are two reasons why in the theoretical work only the simple stratification strategy will be studied. First, no analytical solution could be found for a constant kernel in [Har20] and second, it was evidenced in

[Gar14] and [Gan18] that using order statistics and kernels together can make the calculation of the irradiance estimate from a theoretical/mathematical point of view very difficult.

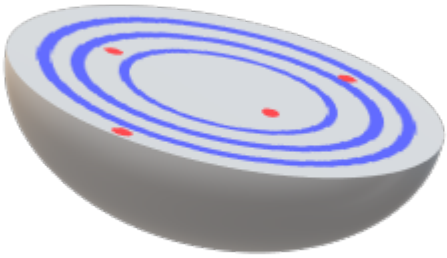


Figure 1.4.: Illustration of the basic stratification strategy.

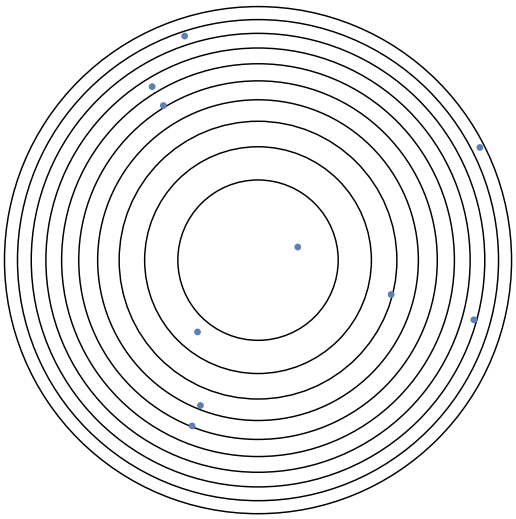


Figure 1.5.: Graphical representation of the basic stratification strategy for 10 rings, programmed in Mathematica.

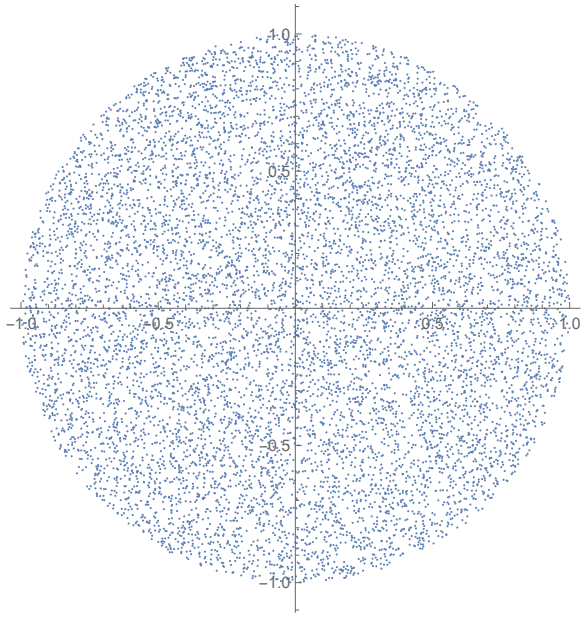


Figure 1.6.: Graphical representation of the photon impacts when using the basic stratification strategy for 10 000 photons, programmed in Mathematica.

1.5.1. Constant kernel

Here the theoretical results obtained in [Har20] about the over- and under-estimation biases for the simple stratification strategy without filtering kernel are explained and presented (for a disk of radius 1 instead of radius $\sqrt{n}$ as in [Har20], due to simplicity). The work in this thesis builds upon these results.
Since using no filtering kernel is equivalent to use a constant kernel, the theoretical study of the stratified Photon Mapping algorithm in [Har20] was done using a constant kernel.
As mentioned before, in this experiment the photons are emitted following the basic stratification strategy under a uniform distribution and the distribution of the emmisive power of the directional light source is also uniform, since both distributions match (both are uniformly distributed and are stratified in the same way) we can assume that all photons have the same flux ϕ. This also means that the incoming flux per unit area is constant at any point in the scene (disk) and therefore the irradiance $I()$ is the same at any point in the scene, including the point P in the center of the disk. This is very convenient because now we know the real value of the irradiance at P, namely $I(P)$, and we can compare it to the irradiance estimate $\hat{I}$ obtained in the Photon Mapping algorithm and see if there is a bias.
Taking into account that the area of the disk is π and that the amount of photons emitted is n, the relation between $I(P)$ and ϕ is the following:

$$\phi = \frac{\pi I(P)}{n} \tag{1.9}$$

Substituting ϕ in the irradiance estimate formula of the Photon Mapping algorithm, we get:

$$\hat{I}(r_k) = \frac{k\phi}{\pi r_k^2} = \frac{kI(P)}{nr_k^2} \tag{1.10}$$

As explained in Section 1.1.3, to obtain an accurate result, we have to iterate the Photon Mapping algorithm several times and then take the average result of the iterations. This can be mathematically modeled by the expected value $E()$, in this context the event of the iterations is the irradiance estimate. To calculate the expected value, we do not only need an event, but also the probability of occurrence of that event and an interval of integration. Since in this experiment the flux is constant, the only variable term in the irradiance estimate formula is r_k (distance from the kth nearest photon impact to the point, where the irradiance estimate is being calculated), this means that the probability of occurrence of a specific value of the irradiance estimate $\hat{I}(r_k)$ is strictly related to r_k. Therefore the probability of occurrence of $\hat{I}(r_k)$ is the probability that the kth nearest photon impact to the point P is at a distance r_k.
The probability distribution function (PDF) of a photon impact to be at a distance d is the normalized circumference formula (normalized means here divided by the area of a ring π/n):

$$f(d) = 2dn \tag{1.11}$$

Now we have a PDF for a photon impact to be at a distance d from P, but what we need is the PDF for the kth nearest photon impact to be at a distance r_k from P, this distribution is given by the kth order statistic $f_{X_{(k)}}(r_k)$ of $f(r_k)$ as in [Gar12]. However due to the stratification strategy used in this experiment the photon impacts are already sorted (because there is only 1 photon per stratum), thus the order statistic of $f_{X_{(k)}}(r_k)$ would be equal to $f(r_k)$ as demonstrated in [Har20].
The interval of integration is given by the radii of the circles that constitute the ring where the kth nearest photon impact to P is located, thus the interval is [t(k-1),t(k)]. The expected value is then:

$$E[\hat{I}(r_k)] = \int_{t(k-1)}^{t(k)} \hat{I}(r_k) f_{X_{(k)}}(r_k) dx = kI(P)\log[\frac{k}{k-1}] \tag{1.12}$$

Since the result of the expected value is not $I(P)$ (real irradiance at P) but $kI(P)\log[\frac{k}{k-1}]$, we can conclude that the bias is the estimation minus the real irradiance ($I(P)$ $(k\log[\frac{k}{k-1}] - 1)$), notice that this only applies to this stratification strategy in particular (basic stratification strategy).
For the 2D constant kernel we get an overestimation bias (graphics for I(P)=1):

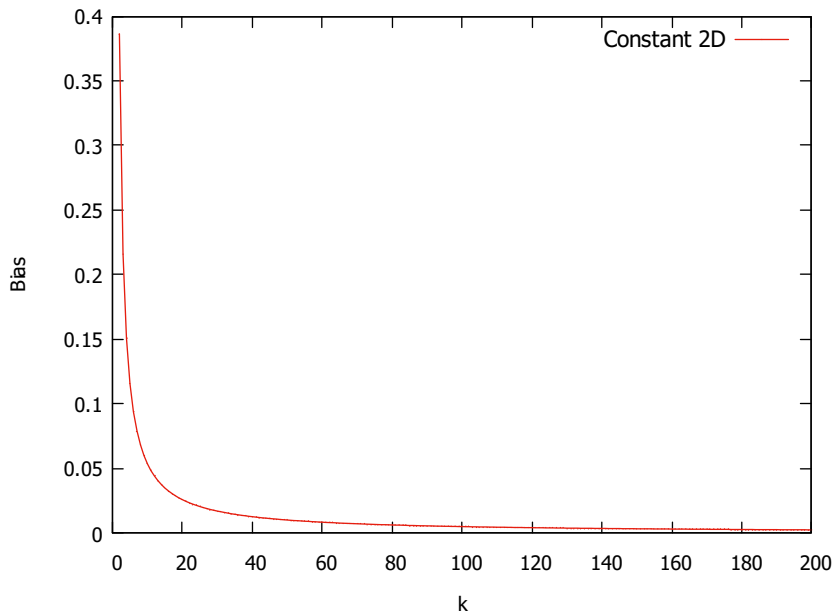


Figure 1.7.: Bias of a Constant kernel 2D.

For the 3D case, we denote the irradiance at the point P in the center of the sphere $PD(P)$ (power density) instead of $I(P)$, also the power density estimate formula $\widehat{PD}(r_k)$ is modified from $\hat{I}(r_k)$ and now it takes into account the volume, and not the area, of the sphere up to the kth nearest photon impact to the center of the sphere. Following the same steps as in the 2D case, we get:

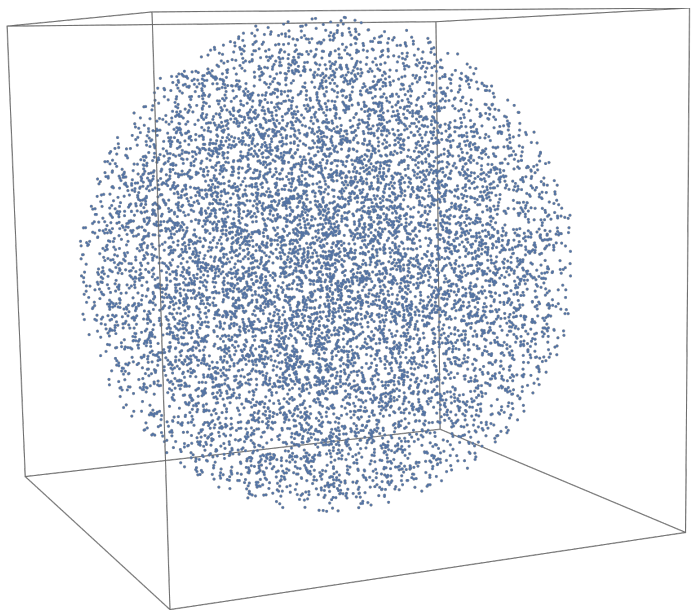


Figure 1.8.: Graphical representation of the position of 10 000 photons when using the basic stratification strategy in 3D, programmed in Mathematica.

$$\phi = \frac{4PD(P)\pi}{3n} \tag{1.13}$$

$$\widehat{PD}(r_k) = \frac{(k)PD(P)}{nr_k^3} \tag{1.14}$$

$$f_{X_{(k)}}(r_k) = 3r_k^2 n \tag{1.15}$$

$$E[\hat{I}(r_k)] = \int_{T(k-1)}^{T(k)} \widehat{PD}(r_k) f_{X_{(k)}}(r_k) dx = (k)PD(P)\log[\frac{k}{k-1}] \tag{1.16}$$

We also get an overestimation bias in 3D (PD(P)=1):

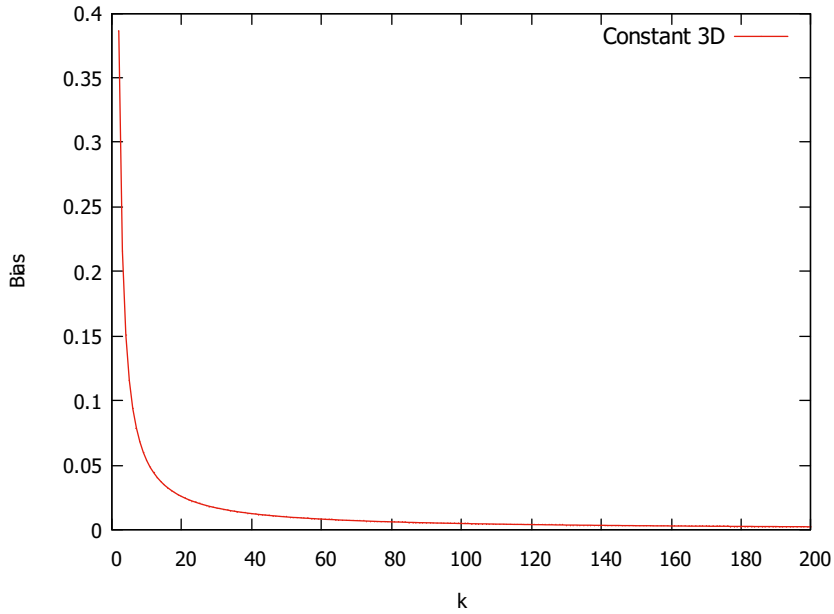


Figure 1.9.: Bias of a Constant kernel 3D.

Notice that in both cases the multiplying term of the biases observed ($k\log[\frac{k}{k-1}]$ for 2D and 3D) can not be eliminated by taking into account only the $(k-1)$ nearest photons since the logarithmic term would not disappear. This means that the strategy proposed in [Gar12] for eliminating the bias does not work in general for the stratified case.

2. Theoretical study

In this section the theoretical study done in [Har20] is extended to non-constant filtering kernels, in concrete, four different kernels are studied, Epanechnikov, Cone, Silverman, and Gaussian kernels. The stratification strategy used in this section is the basic stratification strategy for a disk of radius 1 and a directional light source. The irradiance estimate is calculated for a point P in the center of the disk, also the 3D case for a sphere is studyed.

2.1. Epanechnikov Kernel

Before a kernel can be used in Photon Mapping, it must first be normalized. To derive the normalization constant of a kernel we have to integrate over the area or volume of the group of data points, for which the kernel is being used. Since in this case, the scene is a disk in 2D or a sphere in 3D, it is convenient to use curvilinear coordinates in the integration. This can be done as follows for the 2D Epanechnikov Kernel:

$$C_e = \int_0^{2\pi} \int_0^{r_k} r_i \left(1 - \left(\frac{r_i}{r_k}\right)^2\right) dr_i d\theta = \frac{\pi r_k^2}{2} \tag{2.1}$$

And for the 3D kernel:

$$C_e = \int_0^{2\pi} \int_0^{\pi} \int_0^{r_k} r_i^2 \sin(\phi) \left(1 - \left(\frac{r_i}{r_k}\right)^2\right) dr_i d\phi d\theta = \frac{8\pi r_k^3}{15} \tag{2.2}$$

Notice that the normalization constant depends not only on the dimension of the kernel but also on the size and shape of the group of data points, for which the kernel is being used, a great example of this can be found in [Liu21]. The formula of the normalized Epanechnikov kernel $K_{r_k}(r_i)$ is:

$$K_{r_k}(r_i) = \frac{1}{C_e}\left(1 - (\frac{r_i}{r_k})^2\right) \tag{2.3}$$

The irradiance estimate is:

$$\begin{aligned} \hat{I}(r_1 ... r_k) &= \sum_{i=1}^{k} K_{r_k}(r_i)\phi \\ &= \sum_{i=1}^{k} \frac{2}{\pi r_k^2}\left(1 - (\frac{r_i}{r_k})^2\right)\phi \end{aligned} \tag{2.4}$$

The power density estimate is:

$$\widehat{PD}(r_1 ... r_k) = \sum_{i=1}^{k} \frac{15}{8\pi r_k^3}\left(1 - (\frac{r_i}{r_k})^2\right)\phi \tag{2.5}$$

In mathematics it holds, that the expected value of a summation of several random variables is equal to the summation of their expected values (Equation 1.2), therefore instead of calculating the PDF of the contribution of all photon impacts together to the irradiance estimate, the PDF is calculated for the contribution of each photon impact separately.
The probability distribution function of the contribution of each photon impact separately to the irradiance estimate when using filtering kernels differs from the one obtained before for the constant kernel, the reason for this is that the filtering kernels take into account not only the kth photon impact but also an ith photon impact ($i \leq k$). Since in the basic stratification strategy there is only one photon impact per stratum, the position of the ith nearest photon impact does not depend on the one of the kth nearest photon impact, thus the probability distribution function can be obtained by multiplying the order statistics for both photon impacts (likewise the constant kernel, the PDF here is also the normalized circumference), for the 2D case we get:

$$f_{XY}(r_i, r_k) = f_{X_{(i)}}(r_i) f_{X_{(k)}}(r_k) = 4n^2 r_i r_k \tag{2.6}$$

We calculate the contribution of each photon impact separately as in [Gar14], for $i < k$ we get:

$$\hat{I}_i(r_i, r_k) = \frac{2}{\pi r_k}\left(1 - (\frac{r_i}{r_k})^2\right)\phi \tag{2.7}$$

And for $i = k$ we get:

$$\hat{I}_k(r_k) = 0 \tag{2.8}$$

Since the result of $\hat{I}_k(r_k)$ is 0, the expected value is only to be calculated for $i < k$:

$$\begin{aligned} E[\hat{I}_i] &= \int_{t(k-1)}^{t(k)} \int_{t(i-1)}^{t(i)} \sum_{i=1}^{k-1} \hat{I}_i(r_i, r_k) f_{XY}(r_i, r_k) dr_i dr_k \\ &= \int_{t(k-1)}^{t(k)} \sum_{i=1}^{k-1} \int_{t(i-1)}^{t(i)} \hat{I}_i(r_i, r_k) f_{XY}(r_i, r_k) dr_i dr_k \end{aligned} \tag{2.9}$$

Notice that the expression inside the summation symbol ($\hat{I}_i(r_i, r_k) f_{XY}(r_i, r_k)$) is the same for all values of i, also notice that the only difference in the result comes from the intervals of integration (the rings) and adding all the intervals together constitutes a continuous interval [0,t(k-1)], where 0 corresponds to the lower bound of the first ring and t(k-1) corresponds to the upper bound of the $(k-1)^{th}$ ring. This means that instead of integrating each ring (stratum) separately and then summing them all together, we can integrate all the rings over the interval [0,t(k-1)]. This not only reduces the number of steps in the calculation but also makes the whole calculation easier for computers (in particular for the Cone and Gaussian Kernels, for which the summation of the integrals can not be solved by Mathematica).

$$\begin{aligned} E[\hat{I}_i] &= \int_{t(k-1)}^{t(k)} \int_{0}^{t(k-1)} \hat{I}_i(r_i, r_k) f_{XY}(r_i, r_k) dr_i dr_k \\ &= I(P) \frac{(1-k)(2k\log(k-1) - 2k\log(k) + 1)}{k} \end{aligned} \tag{2.10}$$

The bias for $k \to \infty$ is:

$$\lim_{k\to\infty} I(P)\left(\frac{(1-k)(2k\log(k-1) - 2k\log(k) + 1)}{k} - 1\right) = 0 \tag{2.11}$$

In the following graphic, we observe that the bias for the 2D Epanechnikov kernel is an underestimation bias (I(P)=1).

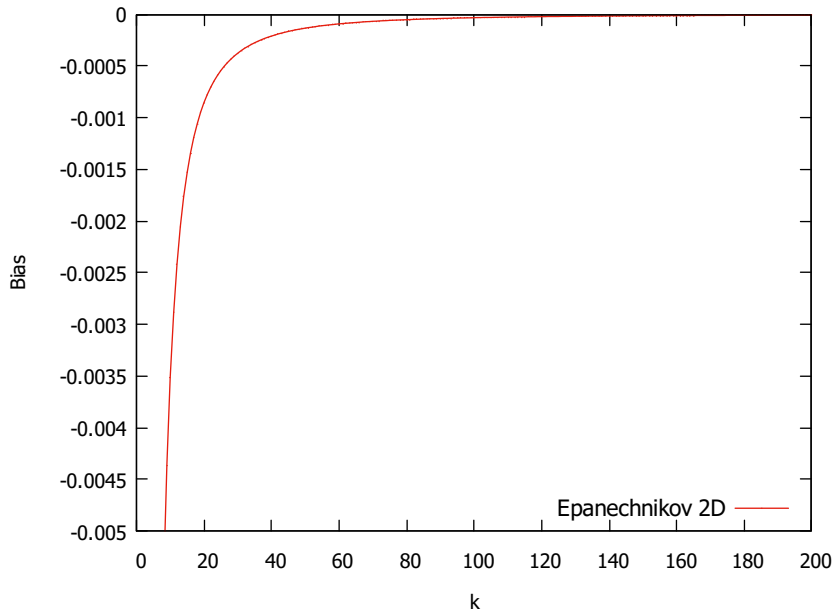


Figure 2.1.: Bias of the Epanechnikov kernel 2D.

Now for the 3D Epanechnikov kernel we get for the PDF:

$$f_{XY}(r_i, r_k) = f_{X_{(i)}}(r_i) f_{X_{(k)}}(r_k) = 9n^2 r_i^2 r_k^2 \tag{2.12}$$

The contributions of $i < k$ and $i = k$ are:

$$\widehat{PD}_i(r_i, r_k) = \frac{15\left(1 - \left(\frac{r_i}{r_k}\right)^2\right)}{8\pi r_k^3}\phi \tag{2.13}$$

$$\widehat{PD}_k(r_k) = 0 \tag{2.14}$$

The expected value is then:

$$\begin{aligned} E[\hat{I}_i] &= \int_{T(k-1)}^{T(k)} \int_0^{T(k-1)} \widehat{PD}_i(r_i, r_k) f_{XY}(r_i, r_k) dr_i dr_k \\ &= PD(P) \frac{(k-1)\left(k^{2/3}(-10\log(k-1) + 10\log(k) - 9) + 9(k-1)^{2/3}\right)}{4k^{2/3}} \end{aligned} \tag{2.15}$$

And the bias for $k \to \infty$ is:

$$\lim_{k \to \infty} PD(P) \left(\frac{(k-1)\left(k^{2/3}(-10\log(k-1) + 10\log(k) - 9) + 9(k-1)^{2/3}\right)}{4k^{2/3}} - 1 \right) = 0 \tag{2.16}$$

For the 3D Epanechnikov kerneL, we get an underestimation bias ($PD(P) = 1$).

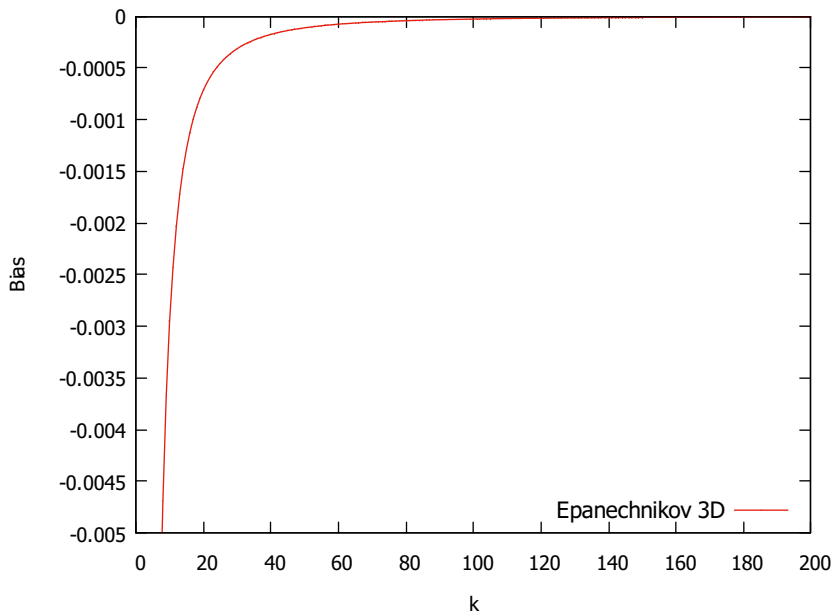


Figure 2.2.: Bias of the Epanechnikov kernel 3D.

2.2. Cone Kernel

The normalization constant for the Cone Kernel in 2D is:

$$C_c = \int_0^{2\pi} \int_0^{r_k} r_i \left(1 - \frac{r_i}{sr_k}\right) dr_i d\theta = 2\pi \left(\frac{r_k^2}{2} - \frac{r_k^2}{3s}\right) \tag{2.17}$$

And for the 3D case:

$$C_c = \int_0^{2\pi} \int_0^{\pi} \int_0^{r_k} r_i^2 \sin(\phi) \left(1 - \frac{r_i}{sr_k}\right) dr_i d\phi d\theta = \frac{\pi(4s-3)r_k^3}{3s} \tag{2.18}$$

The normalized filtering kernel, the irradiance estimate and the power density estimate are then:

$$K_{r_k}(r_i) = \frac{1}{C_c}\left(1 - \frac{r_i}{sr_k}\right) \tag{2.19}$$

$$\hat{I}(r_1 ... r_k) = \sum_{i=1}^{k} \frac{1}{2\pi\left(\frac{r_k^2}{2} - \frac{r_k^2}{3s}\right)} \left(1 - \frac{r_i}{sr_k}\right) \phi \tag{2.20}$$

$$\widehat{PD}(r_1 ... r_k) = \sum_{i=1}^{k} \frac{3s}{\pi(4s-3)r_k^3} \left(1 - \frac{r_i}{sr_k}\right) \phi \tag{2.21}$$

The equations for the contributions $i < k$ and $i = k$ in 2D are:

$$\hat{I}_i(r_i, r_k) = \frac{1 - \frac{r_i}{sr_k}}{\pi r_k^2 (1 - \frac{2}{3s})} \phi \tag{2.22}$$

$$\hat{I}_k(r_k) = \frac{1-\frac{1}{s}}{\pi r_k^2(1-\frac{2}{3s})}\phi \tag{2.23}$$

Expected value of the contribution for $i < k$:

$$\begin{aligned} E[\hat{I}_i] &= \int_{t(k-1)}^{t(k)} \int_0^{t(k-1)} \hat{I}_i(r_i, r_k) f_{XY}(r_i, r_k) dr_i dr_k \\ &= \frac{(k-1)n\phi\left(-3s\log(k-1) + 3s\log(k) + 4\sqrt{\frac{k-1}{k}} - 4\right)}{\pi(3s-2)} \end{aligned} \tag{2.24}$$

As opposed to the Epanechnikov kernel the Cone kernel takes into account the kth photon impact, hence the expected value has to also be calculated for the kth impact, and since $i = k$, $f_{X_{(k)}}$ (PDF used for the constant kernel) will be used instead of f_{XY}:

$$\begin{aligned} E[\hat{I}_k] &= \int_{t(k-1)}^{t(k)} \hat{I}_k(r_k) f_{X_{(k)}}(r_k) dr_k \\ &= \frac{6n\phi\left(\frac{\sqrt{n}r_i}{\sqrt{k-1}} + s\log\left(\frac{\sqrt{k-1}}{\sqrt{n}}\right)\right)}{\pi(2-3s)} - \frac{6n\phi\left(\frac{\sqrt{n}r_i}{\sqrt{k}} + s\log\left(\frac{\sqrt{k}}{\sqrt{n}}\right)\right)}{\pi(2-3s)} \end{aligned} \tag{2.25}$$

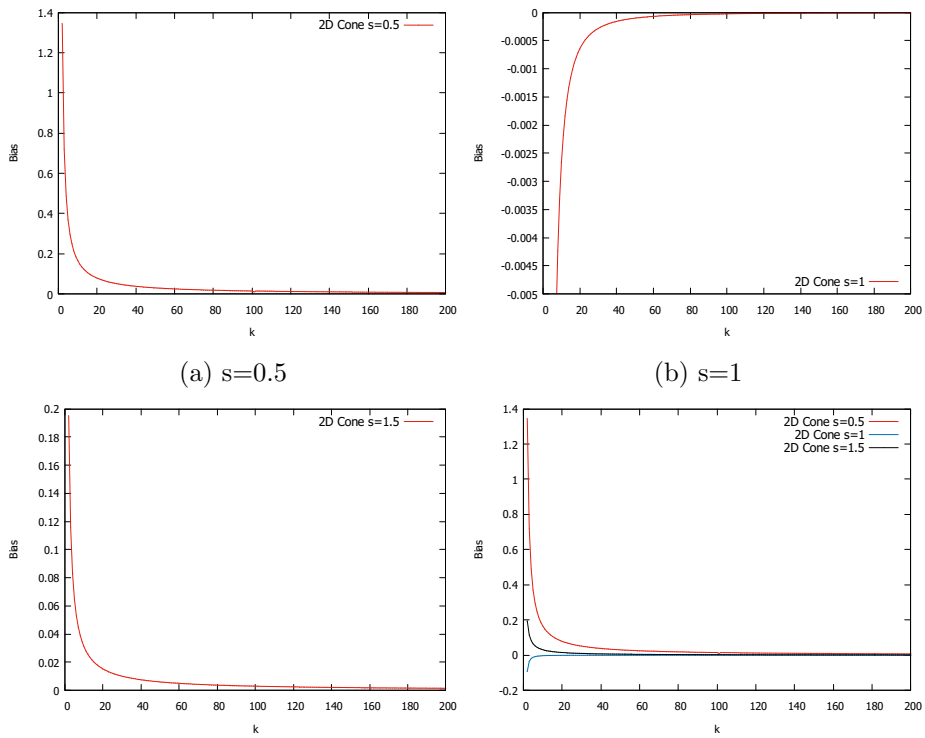
To get the total expected value we add both results:

$$\begin{aligned} E[\hat{I}] &= E[\hat{I}_i] + E[\hat{I}_k] \\ &= I(P)\frac{\frac{4\left(\sqrt{k-1}-\sqrt{k}\right)(k-1)}{\sqrt{k}} + (3-3ks)\log(k-1) + 3(ks-1)\log(k)}{(3s-2)} \end{aligned} \tag{2.26}$$

The bias for $k \to \infty$ is:

$$\lim_{k\to\infty} I(P)\left(\frac{\frac{4\left(\sqrt{k-1}-\sqrt{k}\right)(k-1)}{\sqrt{k}} + (3-3ks)\log(k-1) + 3(ks-1)\log(k)}{(3s-2)} - 1\right) = 0 \tag{2.27}$$

We test the behavior of the bias (assuming $I(P) = 1$) for 3 values of s, namely 0.5, 1 and 1.5, which are standard values of s used in [Gar14]. We observe an overestimation bias for $s = 0.5$ and $s = 1.5$, and an underestimation bias for $s = 1$. Also, the lowest bias is observed for $s = 1$ and the highest for $s = 0.5$ ($I(P) = 1$).



(a) s=0.5 (b) s=1

(c) s=1.5 (d) Comparison

Figure 2.3.: Bias of the Cone kernel 2D.

For the 3D Cone kernel the contributions for $i < k$ and $i = k$ are:

$$\widehat{PD}_i(r_i, r_k) = \frac{1 - \frac{r_i}{sr_k}}{\pi(\frac{4}{3} - \frac{1}{s})r_k^3}\phi \tag{2.28}$$

$$\widehat{PD}_k(r_k) = \frac{\left(1 - \frac{1}{s}\right)\phi}{\pi(\frac{4}{s} - \frac{1}{s})r_k^3} \tag{2.29}$$

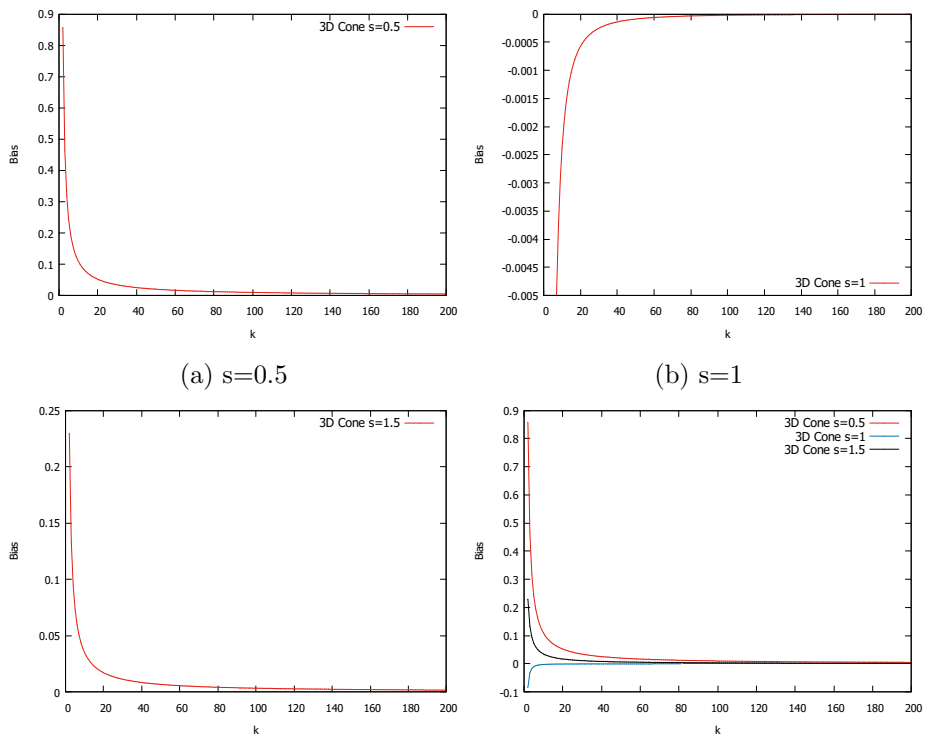
The total expected value is:

$$\begin{aligned} E[\hat{I}] &= E[\hat{I}_i] + E[\hat{I}_k] \\ &= \int_{T(k-1)}^{T(k)} \int_0^{T(k-1)} \widehat{PD}_i(r_i, r_k) f_{XY}(r_i, r_k) dr_i dr_k + \int_{T(k-1)}^{T(k)} \widehat{PD}_k(r_k) f_{X_{(k)}}(r_k) dr_k \\ &= PD(P) \frac{\left((4 - 4ks)\log(k-1) + 4(ks-1)\log(k) + 9\left(\sqrt[3]{\frac{k-1}{k}} - 1\right)(k-1)\right)}{(4s-3)} \end{aligned} \tag{2.30}$$

And the bias for $k \to \infty$ is:

$$\lim_{k\to\infty} PD(P)\left(\frac{\left((4-4ks)\log(k-1)+4(ks-1)\log(k)+9\left(\sqrt[3]{\frac{k-1}{k}}-1\right)(k-1)\right)}{(4s-3)}-1\right)=0 \tag{2.31}$$

Here again we observed an overestimation bias for $s = 0.5$ and $s = 1.5$, whereas for $s = 1$ we observe an underestimation bias. The lowest bias is achieved by $s = 1$ and the highest by $s = 0.5$ ($PD(P) = 1$).



(a) s=0.5 (b) s=1

(c) s=1.5 (d) Comparison

Figure 2.4.: Bias of the Cone kernel 3D.

2.3. Silverman Kernel

The 2D normalization constant for the Silverman Kernel is:

$$C_s = \int_0^{2\pi}\int_0^{r_k} r_i\left(1-\left(\frac{r_i}{r_k}\right)^2\right)^2 dr_i d\theta = \frac{\pi r_k^2}{3} \tag{2.32}$$

3D normalization constant:

$$C_s = \int_0^{2\pi}\int_0^{\pi}\int_0^{r_k} r_i^2\sin(\phi)\left(1-\left(\frac{r_i}{r_k}\right)^2\right)^2 dr_i d\phi d\theta = \frac{32\pi r_k^3}{105} \tag{2.33}$$

The normalized filtering kernel, the irradiance estimate and the power density estimate are:

$$K_{r_k}(r_i) = \frac{1}{C_s}\left(1 - (\frac{r_i}{r_k})^2\right)^2 \tag{2.34}$$

$$\hat{I}(r_1...r_k) = \sum_{i=1}^{k} \frac{3}{\pi r_k^2}\left(1 - (\frac{r_i}{r_k})^2\right)^2 \phi \tag{2.35}$$

$$\widehat{PD}(r_1...r_k) = \sum_{i=1}^{k} \frac{105}{32\pi r_k^3}\left(1 - (\frac{r_i}{r_k})^2\right)^2 \phi \tag{2.36}$$

Likewise the Epanechnikov Kernel the Silverman Kernel just takes into account the contributions up to the $k-1^{th}$ nearest photon impact. The expected value for the 2D case is:

$$\begin{aligned} E[\hat{I}_i(r_k)] &= \int_{t(k-1)}^{t(k)} \int_0^{t(k-1)} \hat{I}_i(r_i, r_k) f_{XY}(r_i, r_k) dr_i dr_k \\ &= I(P) \frac{(1-k)\left(6k^2 \log(k-1) - 6k^2 \log(k) + 4k + 1\right)}{2k^2} \end{aligned} \tag{2.37}$$

The bias for $k \to \infty$ is:

$$\lim_{k\to\infty} I(P)\left(\frac{(1-k)\left(6k^2 \log(k-1) - 6k^2 \log(k) + 4k + 1\right)}{2k^2} - 1\right) = 0 \tag{2.38}$$

The bias for the 2D Silverman kernel is an underestimation bias, here $I(P) = 1$.

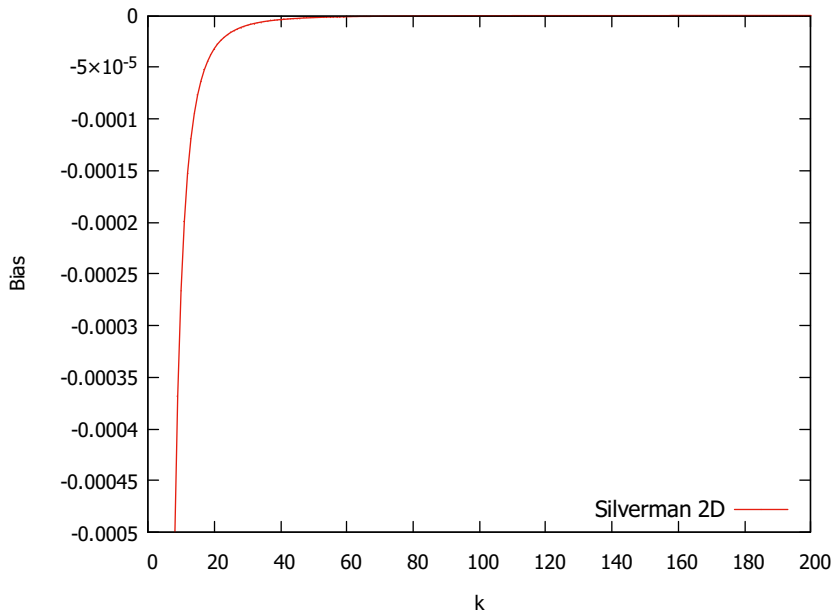


Figure 2.5.: Bias of the Silverman kernel 2D.

Following the same steps for the 3D case, the expected value is:

$$
\begin{aligned}
E[\hat{I}_i(r_k)] &= \int_{T(k-1)}^{T(k)} \int_0^{T(k-1)} \widehat{PD}_i(r_i, r_k) f_{XY}(r_i, r_k) dr_i dr_k \\
&= PD(P) \frac{3}{8}(k-1) \\
&\left(\frac{\frac{3\left(28k^{2/3}\sqrt[3]{k-1}-5k+5\right)\sqrt[3]{k-1}}{k^{4/3}} + 140\log\left(\sqrt[3]{k}\right) - 140\log\left(\sqrt[3]{k-1}\right) - 69}{4} \right)
\end{aligned}
\tag{2.39}
$$

The bias of the 3D Silverman kernel for $k \to \infty$ is also 0 and it is an underestimation bias (graphics with $(PD(P) = 1)$).

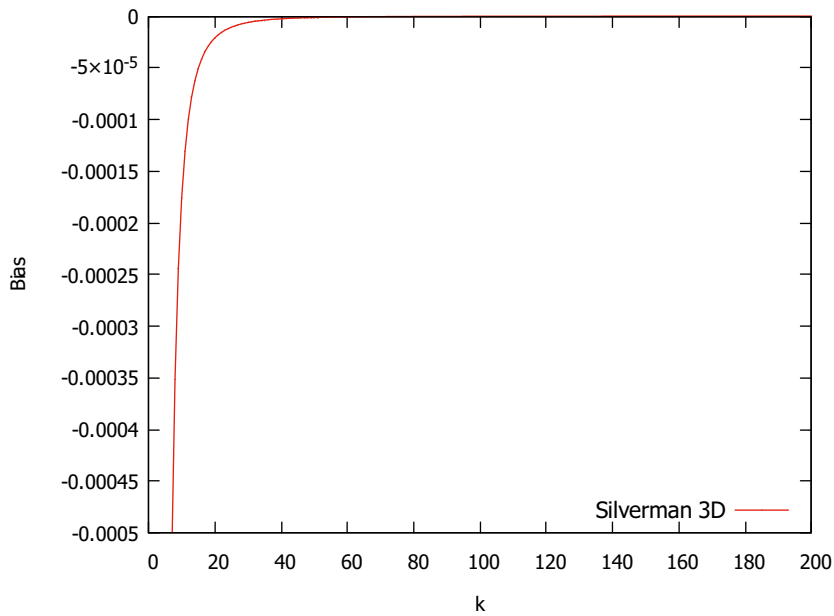


Figure 2.6.: Bias of the Silverman kernel 3D.

2.4. Gaussian Kernel

The 2D normalization constant of the Gaussian Kernel is:

$$
C_g = \int_0^{2\pi} \int_0^{r_k} r_i \left(1 - \frac{1 - e^{-\frac{\beta r_i^2}{2r_k^2}}}{1 - e^{-\beta}} \right) dr_i d\theta = \frac{\pi\left(-\beta - 2e^{\beta/2} + 2e^{\beta}\right) r_k^2}{\left(e^{\beta} - 1\right)\beta} \tag{2.40}
$$

3D normalization constant:

$$\begin{aligned} C_g &= \int_0^{2\pi}\int_0^{\pi}\int_0^{r_k} r_i^2 \sin(\phi)\left(1-\frac{1-e^{-\frac{\beta r_i^2}{2r_k^2}}}{1-e^{-\beta}}\right) dr_i d\phi d\theta \\ &= \frac{2\pi\left(3\sqrt{2\pi}e^{\beta}\mathrm{erf}\left(\frac{\sqrt{\beta}}{\sqrt{2}}\right)-2\sqrt{\beta}\left(\beta+3e^{\beta/2}\right)\right)r_k^3}{3\left(e^{\beta}-1\right)\beta^{3/2}} \end{aligned} \tag{2.41}$$

Where the Gaussian Error Function is defined as follows:

$$erf(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}dt \tag{2.42}$$

The normalized filtering kernel, the irradiance estimate and the power density estimate are:

$$K_{r_k}(r_i) = \frac{1}{C_g}\left(1-\frac{1-e^{-\frac{\beta r_i^2}{2r_k^2}}}{1-e^{-\beta}}\right) \tag{2.43}$$

$$\hat{I}(r_1...r_k) = \sum_{i=1}^{k}\frac{\alpha}{r_k^2}\left(1-\frac{1-e^{-\frac{\beta r_i^2}{2r_k^2}}}{1-e^{-\beta}}\right)\phi \tag{2.44}$$

$$\widehat{PD}(r_1...r_k) = \sum_{i=1}^{k}\frac{\alpha_v}{r_k^3}\left(1-\frac{1-e^{-\frac{\beta r_i^2}{2r_k^2}}}{1-e^{-\beta}}\right)\phi \tag{2.45}$$

Where α and α_v are defined as follows:

$$\alpha = \frac{\left(e^{\beta}-1\right)\beta}{-\beta-2e^{\beta/2}+2e^{\beta}} = 1.728 \tag{2.46}$$

$$\alpha_v = \frac{2\left(e^{\beta}-1\right)\beta^{3/2}}{3\sqrt{2\pi}e^{\beta}\mathrm{erf}\left(\frac{\sqrt{\beta}}{\sqrt{2}}\right)-2\sqrt{\beta}\left(\beta+3e^{\beta/2}\right)} = 1.97758 \tag{2.47}$$

For the Gaussian kernel we need to calculate the expected value for $i < k$ and $i = k$ separately, as done before for the Cone Kernel. For the 2D case we get:

$$\begin{aligned} E[\hat{I}] &= E[\hat{I}_i] + E[\hat{I}_k] \\ &= \int_{t(k-1)}^{t(k)}\int_0^{t(k-1)} \hat{I}_i(r_i,r_k) f_{XY}(r_i,r_k) dr_i dr_k + \int_{t(k-1)}^{t(k)} \hat{I}_k(r_k) f_{X_{(k)}}(r_k) dr_k \\ &= I(P)\mathrm{U} \end{aligned} \tag{2.48}$$

And U is:

$$\begin{aligned} U = &\frac{-\alpha\left(\left(-2e^{\beta/2}\left(e^{\beta/2}+k\left(-e^{\frac{\beta}{2k}}\right)+k-1\right)\right)+\beta\left(e^{\beta/2}-k\right)\left(\log(k-1)-\log(k)\right)\right)}{\left(e^{\beta}-1\right)\beta} \\ &-\alpha\frac{e^{\beta}\beta(k-1)\left(E_1\left(\frac{\beta}{2}\right)-E_1\left(\frac{\beta(k-1)}{2k}\right)\right)}{\left(e^{\beta}-1\right)\beta} \end{aligned} \tag{2.49}$$

Where the Exponential Integral Function for complex values is defined as follows:

$$E_1(z) = \int_z^{\infty} \frac{e^{-t}}{t} dt \tag{2.50}$$

The bias of the 2D Gaussian kernel for $k \to \infty$ is also 0 and it is an overestimation bias (graphics with $I(P) = 1$).

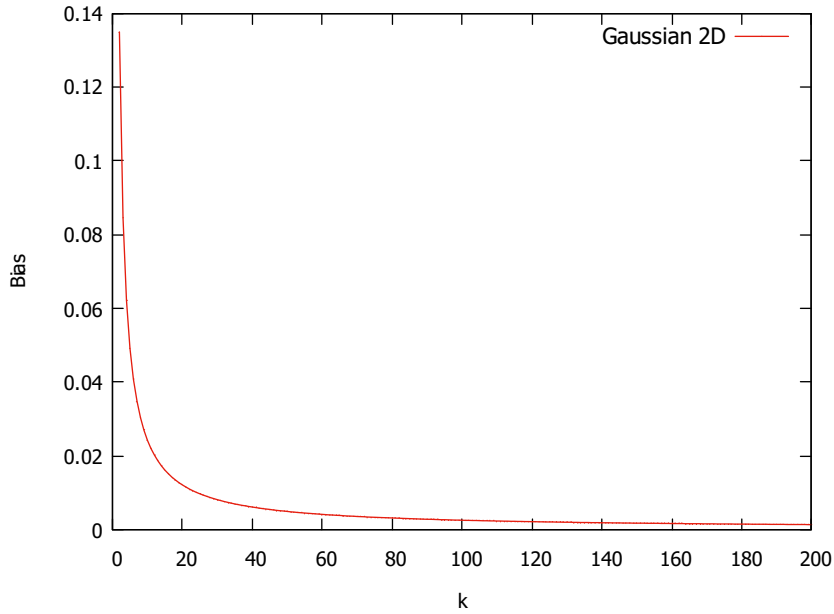


Figure 2.7.: Bias of the Gaussian kernel 2D.

For the 3D case:

$$\begin{aligned} E[\hat{I}] &= E[\hat{I}_i] + E[\hat{I}_k] \\ &= \int_{T(k-1)}^{T(k)} \int_0^{T(k-1)} \widehat{PD}_i(r_i, r_k) f_{XY}(r_i, r_k) dr_i dr_k + \int_{T(k-1)}^{T(k)} \widehat{PD}_k(r_k) f_{X_{(k)}}(r_k) dr_k \\ &= PD(P)\mathrm{V} \end{aligned} \tag{2.51}$$

V is:

$$
\begin{aligned}
V = & \frac{3\alpha_v\left(2\sqrt{2\pi}e^{\beta} - 4\sqrt{\beta}k^{2/3}\sqrt[3]{k-1}e^{\beta-\frac{1}{2}\beta\left(\frac{k-1}{k}\right)^{2/3}} + 4e^{\beta/2}\sqrt{\beta}(k-1)\right)}{4\left(e^{\beta}-1\right)\beta^{3/2}} \\
& + \frac{3\alpha_v\left(2\sqrt{2\pi}e^{\beta}(k-1)\mathrm{erfc}\left(\frac{\sqrt{\beta}}{\sqrt{2}}\right) - 2\sqrt{2\pi}e^{\beta}(k)\mathrm{erfc}\left(\frac{\sqrt{\beta}\sqrt[3]{\frac{k-1}{k}}}{\sqrt{2}}\right)\right)}{4\left(e^{\beta}-1\right)\beta^{3/2}} \\
& + \frac{9e^{\beta}(1-k)\alpha_v E_1\left(\frac{\beta}{2}\right) - E_1\left(\frac{1}{2}\left(\frac{k-1}{k}\right)^{2/3}\beta\right)}{4\left(e^{\beta}-1\right)} \\
& + \frac{3e^{\beta}(1-k)\alpha_v \mathrm{Ei}\left(-\frac{\beta}{2}\right) - \mathrm{Ei}\left(-\frac{1}{2}\left(\frac{k-1}{k}\right)^{2/3}\beta\right)}{4\left(e^{\beta}-1\right)} \\
& - \frac{\left(4e^{\beta/2}+4k\right)\left(\log(k-1)-\log(k)\right)\alpha_v}{4\left(e^{\beta}-1\right)}
\end{aligned}
\tag{2.52}
$$

Where the Complementary Error Function ($erfc$) and the Exponential Integral Function for real non-zero numbers (Ei) are defined as follows:

$$erfc(z) = 1 - erf(z) \tag{2.53}$$

$$Ei(x) = -\int_{-x}^{\infty} \frac{e^{-t}}{t} dt \tag{2.54}$$

The bias of the 3D Gaussian kernel for $k \to \infty$ is also 0 and it is an overestimation bias (graphics with $PD(P) = 1$).

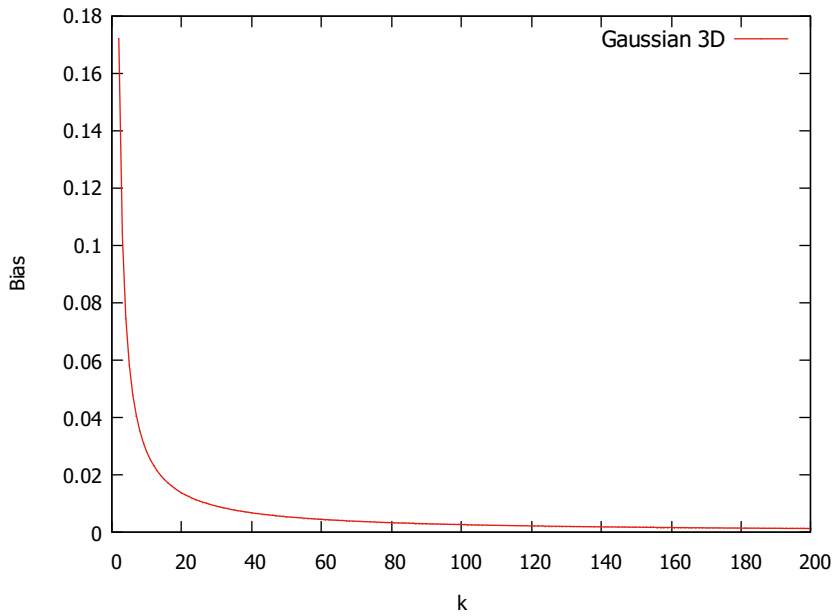


Figure 2.8.: Bias of the Gaussian kernel 3D.

When analyzing the theoretical results obtained here and comparing them to the results obtained in [Gar12] we observe that in all cases the bias disappears when $k \to \infty$, being this the true source of the bias since in practice the Photon Mapping algorithm can not be performed for an infinite number $n > k$ of photons, notice that the relation $n > k$ means that although k and n are infinite, the infinity of n has to be greater. This was also mathematically described for a more general case by Jensen in [Jen02]. As opposed to the results obtained in [Gar14] for random sampling the four kernels studied in this work can not eliminate the bias for the basic stratification strategy.

3. Empirical study

In this section, the Photon Mapping algorithm is empirically studied. First the over- and under-estimation biases are empirically studied, then in the second part, the effects of different stratification strategies on the standard deviation of the irradiance estimate are studied in a real scene (Cornell Box).

3.1. Over- and under-estimation biases

To empirically test the results obtained in the theoretical study, the same experiment with basic stratification and a directional light source is implemented in a C program, in this program the relative error of the irradiance estimate at the point P in the center of the disk (and sphere) is calculated for different values of k. the results are presented in graphics consisting of the relative error on the y-axis and k on the x-axis. For each kernel, the experiment is performed for 100 000 photons and an interval [2,200] of k values, these are standard parameters used in previous works on theoretical Photon Mapping. To simulate the stratification on the scene two methods are studied, rejection sampling and a method that makes use of polar and spherical coordinates as described in [Art15].
Rejection sampling is known to be faster in generating uniformly distributed random points on a disk or within a sphere, however, this assumption does not hold for the stratification used in this experiment. This happens because in random sampling the probability of picking the right point (a point inside the disk or within the sphere) each iteration is very high, about 4/5 for the disk and about 1/2 for the sphere. however for a disk divided in n rings, whose areas are n times smaller than the area of the disk, the probability, when picking one point per ring, can dramatically decrease depending on the implementation, in a worst-case scenario (applying rejection sampling on the whole disk and picking the points ring by ring) this probability could be 0.0007 for $n = 100\ 000$. Therefore in this work, the method used to simulate the stratification on the scene is the one using polar and spherical coordinates. One of the reasons why rejection sampling is faster in the unstratified case is that using polar or spherical coordinates requires to use of costly trigonometrical functions such as cos or sin several times. However since in this experiment the photons are emitted from a directional light source, they fall down into the disk keeping the same distance to the center of the disk, hence we only need the distances of the photon impacts to the center of the disk and not their positions. Therefore we can simplify the algorithm to only one expression and by doing so one also avoids calling several times the function for generating random numbers or trigonometrical functions, the 2D distance formula can be derived as follows:

Algorithm 1 Ring Point Picking algorithm

Input: $r_{\mathbf{outer}}, r_{\mathbf{inner}}$

$\theta \leftarrow \text{rand}()2\pi$
$r \leftarrow \text{rand}() \left(r_{\text{outer}}^2 - r_{\text{inner}}^2\right) + r_{\text{inner}}^2$
$x \leftarrow \sqrt{r}\cos(\theta)$
$y \leftarrow \sqrt{r}\sin(\theta)$
return (x, y)

Figure 3.1.: Ring Point Picking algorithm, modification of the Direct Disk Point Picking method described in [Art15].

Where rand() is a function that generates a uniformly distributed random number between 0 and 1, the radius r and the angle θ are the polar coordinates of the photon, and r_{inner} and r_{outer} are the radii of the circles that build a ring.
Since in this experiment, the coordinates of the points are not needed but their distances to the center, we can reduce the algorithm to one single formula.

$$\begin{aligned} Distance &= \sqrt{(r\sin(\theta))^2 + (r\cos(\theta))^2} \\ &= \sqrt{r\left(\sin^2(\theta) + \cos^2(\theta)\right)} \\ &= \sqrt{r} \\ &= \sqrt{\text{rand}()\left(r_{\text{outer}}^2 - r_{\text{inner}}^2\right) + r_{\text{inner}}^2} \end{aligned} \tag{3.1}$$

The formula obtained above was also described in [Sch21]. From the theoretical part of this work we know $r_{\text{outer}} = \frac{\sqrt{i}}{\sqrt{n}}$ and $r_{\text{inner}} = \frac{\sqrt{i-1}}{\sqrt{n}}$, where i is a ring in the disk and n is the total amount of photon impacts. Substituting, we get:

$$\sqrt{\left(\left(\frac{\sqrt{i}}{\sqrt{n}}\right)^2 - \left(\frac{\sqrt{i-1}}{\sqrt{n}}\right)^2\right)\text{rand}() + \left(\frac{\sqrt{i-1}}{\sqrt{n}}\right)^2} = \sqrt{\left(\frac{i}{n} - \frac{i-1}{n}\right)\text{rand}() + \frac{i-1}{n}} \tag{3.2}$$

And therefore the simplified 2D distance function is:

$$Distance(i, n) = \sqrt{(i - 1 + rand())/n} \tag{3.3}$$

For the 3D case the function can be obtained analogously:

$$Distance(i, n) = \sqrt[3]{(i - 1 + rand())/n} \tag{3.4}$$

Another optimization that can be made is in the way that the irradiance estimate for each kernel is calculated, in this work the formulas for calculating the irradiance estimate were modified taking advantage of the distributive property of the multiplication over addition, the goal here was to reduce the amount of calculation to be performed within a loop. The big square brackets notation is used here to indicate that, to achieve the maximal speedup, the

terms outside these brackets have to be multiplied to the final result only after all iterations are done.
The modified irradiance estimate formula for the 2D Epanechnikov kernel can be derived as follows:

$$\begin{aligned}
Epanechnikov2D(r_i, r_k) &= \sum_{i=1}^{k} \frac{2\phi\left(1-\left(\frac{r_i}{r_k}\right)^2\right)}{\pi r_k^2} \\
&= \frac{2\phi \sum_{i=1}^{k} \frac{1-\left(\frac{r_i}{r_k}\right)^2}{r_k^2}}{\pi} \\
&= \frac{2\phi \sum_{i=1}^{k} \frac{r_k^2-r_i^2}{r_k^2 r_k^2}}{\pi} \\
&= \frac{2\phi\left(\sum_{i=1}^{k} r_k^2 - \sum_{i=1}^{k} r_i^2\right)}{\pi r_k^4} \\
&= \frac{2\phi\left(k r_k^2 - \sum_{i=1}^{k} r_i^2\right)}{\pi r_k^4} \\
&= \frac{2\phi\left(-\sum_{i=1}^{k-1} r_i^2 + k r_k^2 - r_k^2\right)}{\pi r_k^4} \\
&= \frac{2\phi}{\pi}\left[\frac{(k-1)r_k^2 - \sum_{i=1}^{k-1} r_i^2}{r_k^4}\right]
\end{aligned} \tag{3.5}$$

One can also derive the optimization for the 3D Epanechnikov kernel following the same steps, notice that the only changes in the final result come from the normalization constant:

$$Epanechnikov3D(r_i, r_k) = \frac{15\phi}{8\pi}\left[\frac{(k-1)r_k^2 - \sum_{i=1}^{k-1} r_i^2}{r_k^5}\right] \tag{3.6}$$

The optimization for the other kernels can be derived analogously, the optimization of the irradiance estimate for the Cone kernel is:

$$Cone2D(r_i, r_k) = \frac{\phi}{\pi\left(s-\frac{2}{3}\right)}\left[\frac{(ks-1)r_k - \sum_{i=1}^{k-1} r_i}{r_k^3}\right] \tag{3.7}$$

$$Cone3D(r_i, r_k) = \frac{3s\phi}{\pi(4s-3)s}\left[\frac{r_k(ks-1) - \sum_{i=1}^{k-1} r_i}{r_k^4}\right] \tag{3.8}$$

The optimization for the Gaussian kernel is:

$$Gaussian2D(r_i, r_k) = \frac{\alpha\phi}{\pi(1-e^{-\beta})}\left[\frac{\sum_{i=1}^{k} e^{-\frac{\beta r_i^2}{2r_k^2}} - e^{-\beta}k}{r_k^2}\right] \tag{3.9}$$

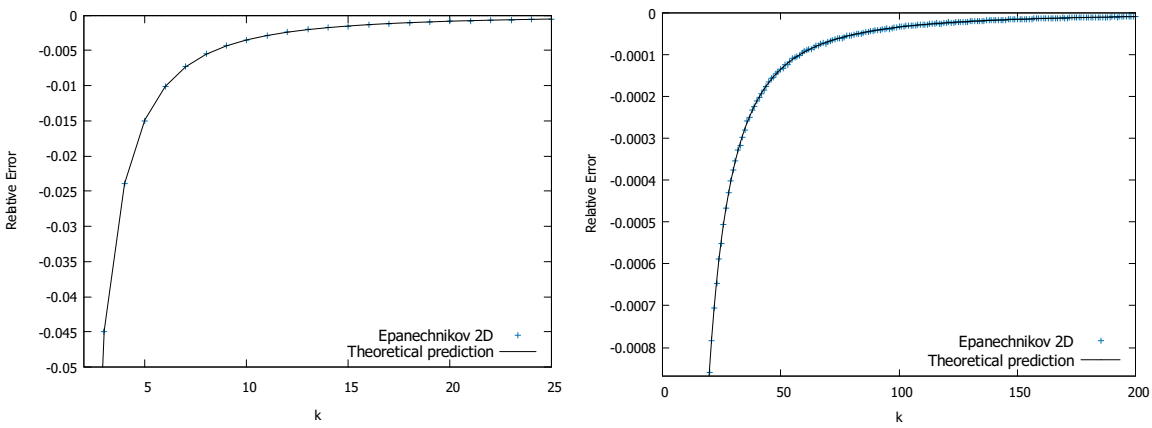
$$Gaussian3D(r_i, r_k) = \frac{3\alpha_v\phi}{4\pi(1-e^{-\beta})}\left[\frac{\sum_{i=1}^{k} e^{-\frac{\beta r_i^2}{2r_k^2}} - e^{-\beta}k}{r_k^3}\right] \tag{3.10}$$

And for the Silverman kernel the optimization is:

$$Silverman2D(r_i, r_k) = \frac{3\phi}{\pi}\left[\frac{-2r_k^2\left(\sum_{i=1}^{k} r_i^2\right) + \sum_{i=1}^{k-1} r_i^4 + (k+1)r_k^4}{r_k^6}\right] \tag{3.11}$$

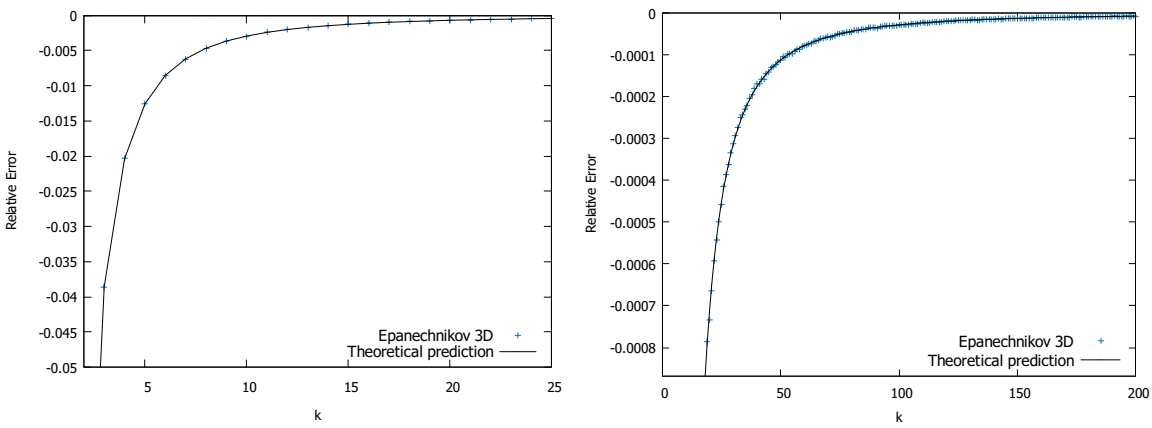
$$Silverman3D(r_i, r_k) = \frac{105\phi}{32\pi}\left[\frac{-2r_k^2\left(\sum_{i=1}^{k} r_i^2\right) + \sum_{i=1}^{k-1} r_i^4 + (k+1)r_k^4}{r_k^7}\right] \tag{3.12}$$

For the empirical work on the over- and under-estimation biases in the 2D and 3D cases, 1 000 0000 iterations were performed for all kernels, however, it was empirically observed that for the Epanechnikov kernel around 10 000 iterations were enough to eliminate most of the noise:



(a) k [2-25] (b) k [2-200]

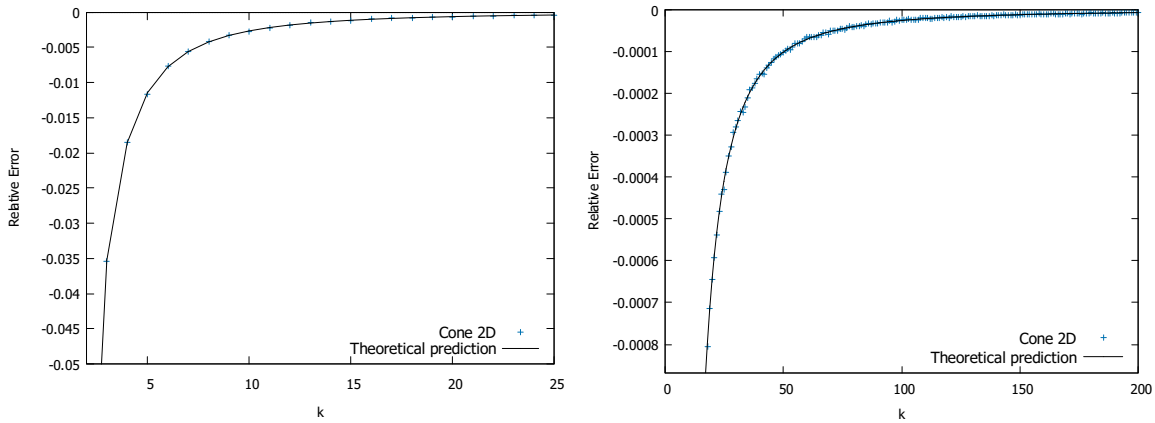
Figure 3.2.: Simulation of the relative error for the Epanechnikov kernel 2D



(a) k [2-25] (b) k [2-200]

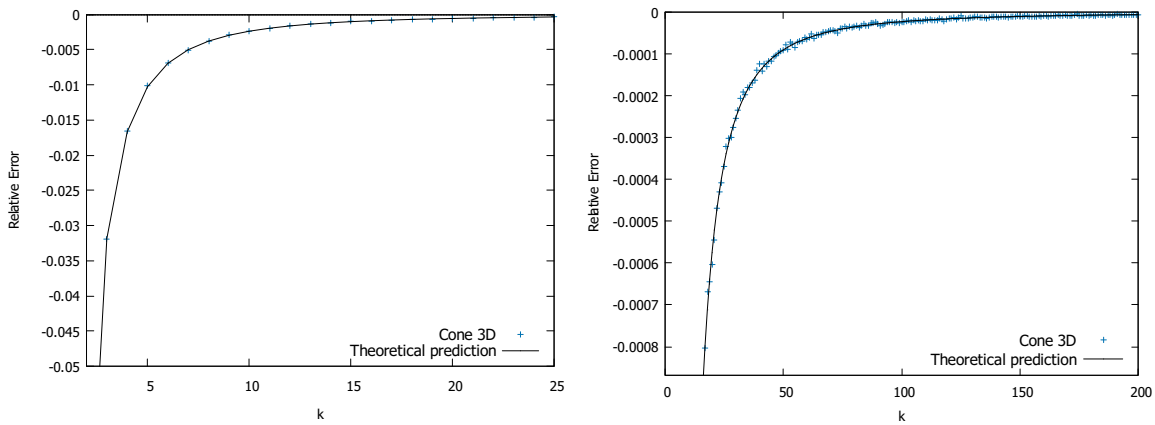
Figure 3.3.: Simulation of the relative error for the Epanechnikov kernel 3D

For the Cone kernel was empirically observed that between 20 000 to 30 000 iterations are enough to obtain an accurate result in the 2D case and for the 3D case about 100 000 iterations are enough to eliminate most of the noise. The calculations for the Cone kernel were made for $s = 1$ because in the theoretical study this value of the slope showed the lowest bias:



(a) k [2-25] (b) k [2-200]

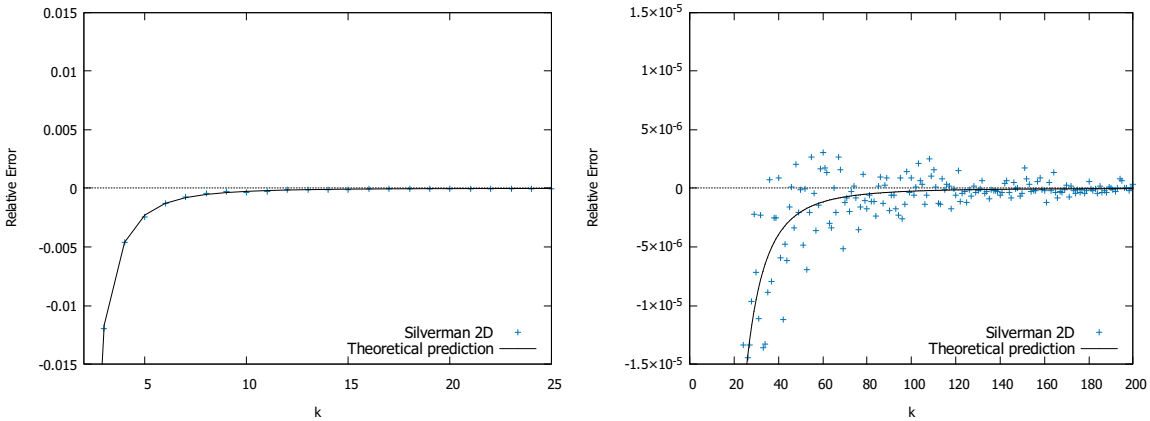
Figure 3.4.: Simulation of the relative error for the Cone kernel 2D



(a) k [2-25] (b) k [2-200]

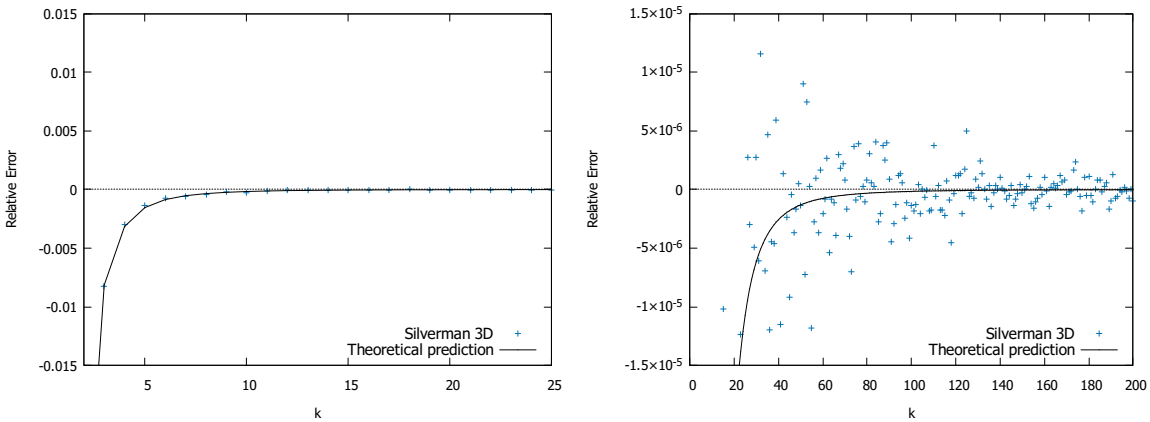
Figure 3.5.: Simulation of the relative error for the Cone kernel 3D

For the 2D Silverman kernel about 50 million iterations are needed to reduce most of the noise in the result and 150 million iterations for the 3D case:



(a) k [2-25] (b) k [2-200]

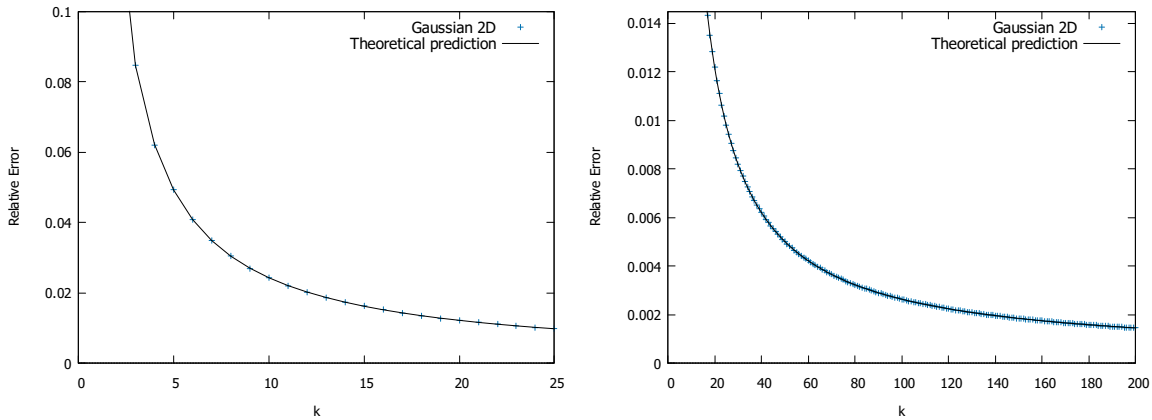
Figure 3.6.: Simulation of the relative error for the Silverman kernel 2D



(a) k [2-25] (b) k [2-200]

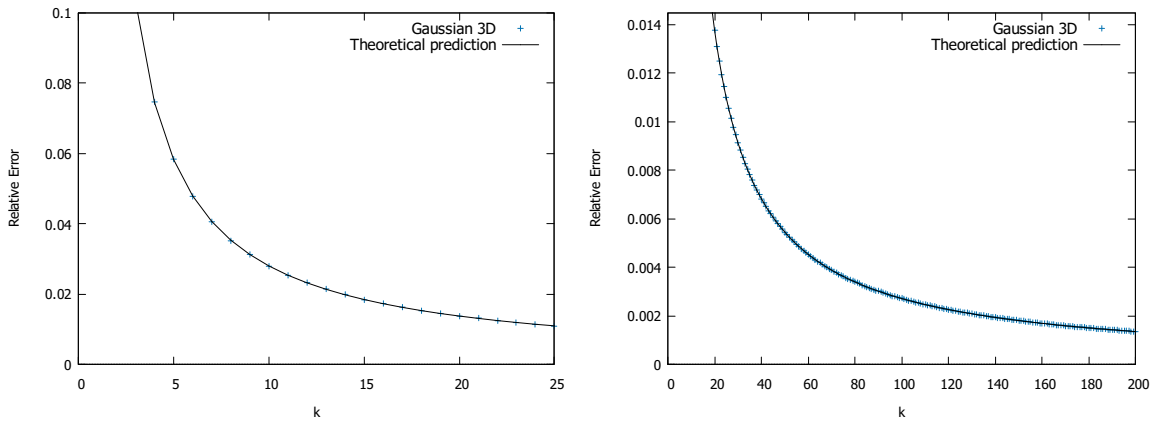
Figure 3.7.: Simulation of the relative error for the Silverman kernel 3D

For the empirical implementation of the Gaussian kernel only 1000 iterations are needed to reduce most of the noise:



(a) k [2-25] (b) k [2-200]

Figure 3.8.: Simulation of the relative error for the Gaussian kernel 2D



(a) k [2-25] (b) k [2-200]

Figure 3.9.: Simulation of the relative error for the Gaussian kernel 3D

Here can be observed that all of the studied kernels showed lower bias than the constant kernel implementation, it was also observed that among the kernels the highest accuracy (lowest bias) was obtained by the Silverman kernel and the least accurate kernel was the Gaussian kernel. These graphics also show that the more accuracy obtained by a kernel the more iterations were needed to reduce the noise.

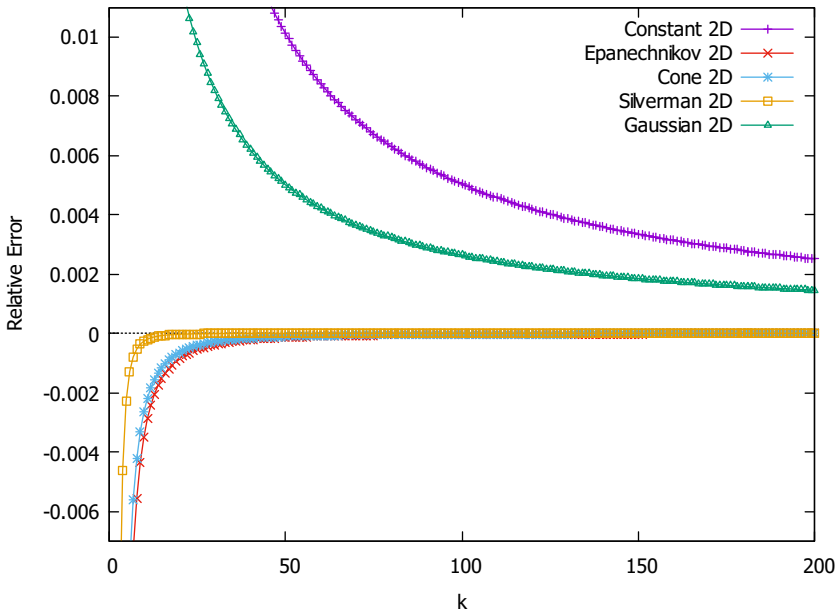


Figure 3.10.: Simulation of the relative error for the 4 kernels 2D.

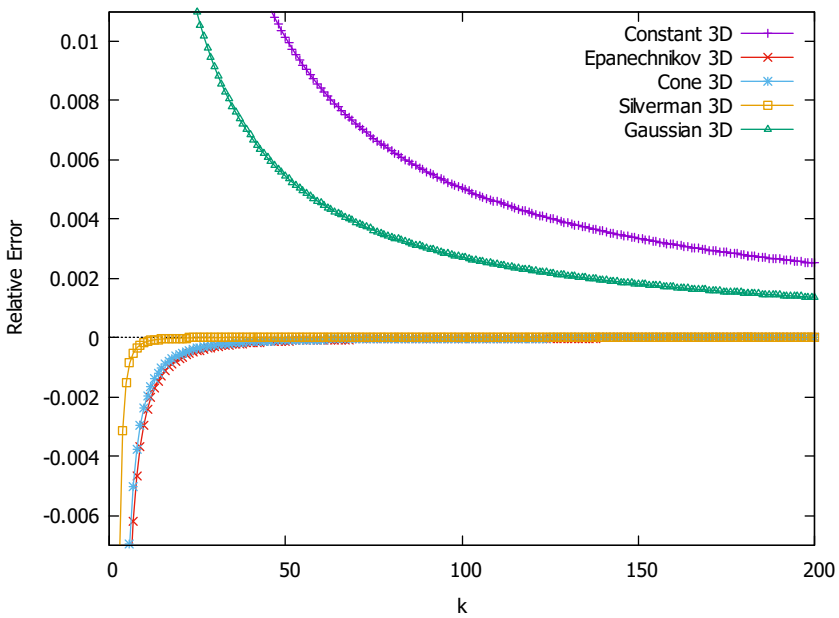


Figure 3.11.: Simulation of the relative error for the 4 kernels 3D.

Since the only difference between the 2D and 3D case in the formula for calculating the irradiance estimate is the normalization constant of each kernel, we observe no significant difference between the two graphics (2D and 3D) as expected, notice that in the two graphics above (Figures 3.10 and 3.11) the kernels stay in the same order with respect to the relative error as in [Gar14].

3.2. Cornell Box

To study the behavior of the Photon Mapping algorithm with stratification in a real scene, three different stratifications will be implemented in the same scene. The scene consists of a room with a floating box inside illuminated by a square light source (square lamp). The original C++ code for random sampling and a constant kernel was provided by Dr. García. The author of this thesis implemented in that code the methods for the different stratification strategies and kernels.
In a square light source the positions of emission of the photons are uniformly distributed in the square and the directions of emission are distributed following a cosine distribution on a half-sphere, the cosine distribution on the half-sphere can be obtained by uniformly distributing the photons on a disk and then projecting them to the half-sphere, this guarantees a cosine distribution on the surface of the half-sphere. As a consequence of this, the photons do not fall down into the scene in the same position, in which they were emitted. In this work, the light source is stratified knowing that it will not necessarily translate into the same stratification in the scene. However, at least a better distribution of the photons throughout the scene is expected due to the stratification of the light source. Since the stratification achieved in the scene is most likely not the same as in the light source, there is no point in emitting only one photon per stratum, instead, a number x of photons is emitted per stratum.
First, the position of emission of photons is stratified, to do so the square lamp is divided into 100 squares of same area, in each square 1200 photons are emitted (Figure 3.13). In the second stratification strategy, the directions in which the photons are emitted are stratified, the photons are here organized following one of the stratifications used in [Har20] (the so-called stratification with order statistics), which consists of a disk divided into smaller rings that at the same time are subdivided into smaller strata for a total of 120 strata and 1 000 photons per stratum (Figure 3.14), in this work this stratification strategy is referred to as the disk strategy. Finally, the third strategy combines both strategies mentioned before, in the three strategies 120 000 photons are emitted.
The purpose here is to generate a secondary stratification on the scene to analyze if the stratification used in the theoretical part of this work and therefore in [Har20] also finds application on a real scene and in particular if it does reduce the variance of the Photon Mapping algorithm in the scene.
The illustration in Figure 3.12 represents the scene on which the Photon Mapping algorithm is used, the red dot in the center of the scene represents the origin at (0,0,0), for which the irradiance is being estimated. since the irradiance estimate is only being calculated for one point in the scene, the photon map data structure is not implemented, to reduce the computation time of the program.

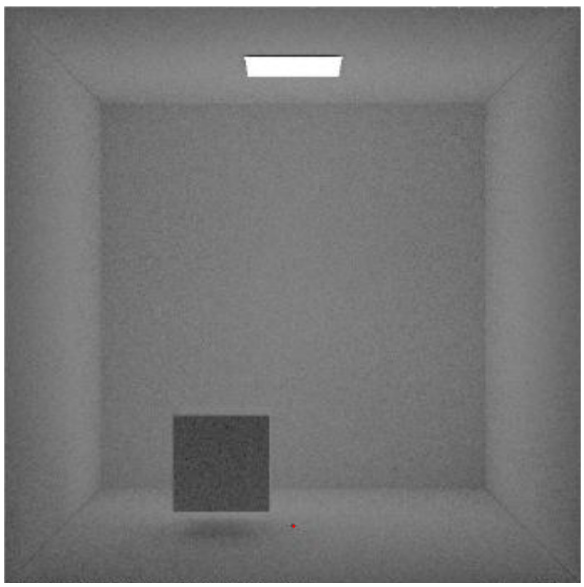


Figure 3.12.: Illustration of Cornell Box scene.

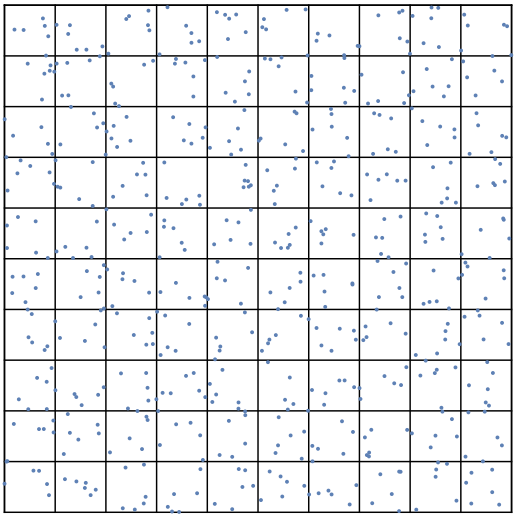


Figure 3.13.: Representation of square strategy with 6 photons per stratum, programmed in C++ and Mahtematica.

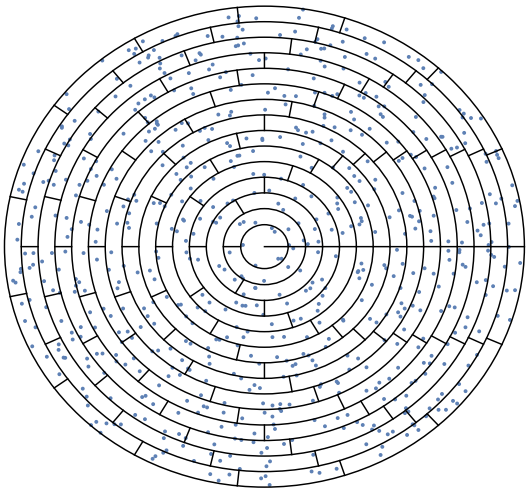


Figure 3.14.: Representation of disk strategy with 6 photons per stratum, programmed in C++ and Mahtematica.

To obtain an average result for all three stratification strategies 10 000 iterations were performed and since this algorithm can be very computationally expensive, only the perfect squares from 2 to 15 were used as values for k. In a previous work on this topic, an optimization on how to generate the disk for the stratified emission of photons was introduced [Sch21], this optimization was based on the fact that one can partition an integral into many pieces without modifying the result and consisted of eliminating the inner subdivisions of the rings. However, the speedup gained with this strategy is only due to calling fewer times the rand() function, for this reason, the author of this work stuck to the original stratification strategy used in [Har20].

In the following graphics, the kernels are analyzed individually (for the Cone kernel $s = 1$) and the different strategies compared to each other. Notice that although the purpose of this work is to study the variance in a real case scenario, the standard deviation is studied instead, this is due to the small size of the values result of the calculations.

For all four kernels, it is observed that both the disk stratification strategy and the combined stratification strategy show a reduction in the standard deviation compared to the unstratified case as expected, also a minor increase and therefore a worsening in the standard deviation is observed for the kernels in the square stratification strategy.

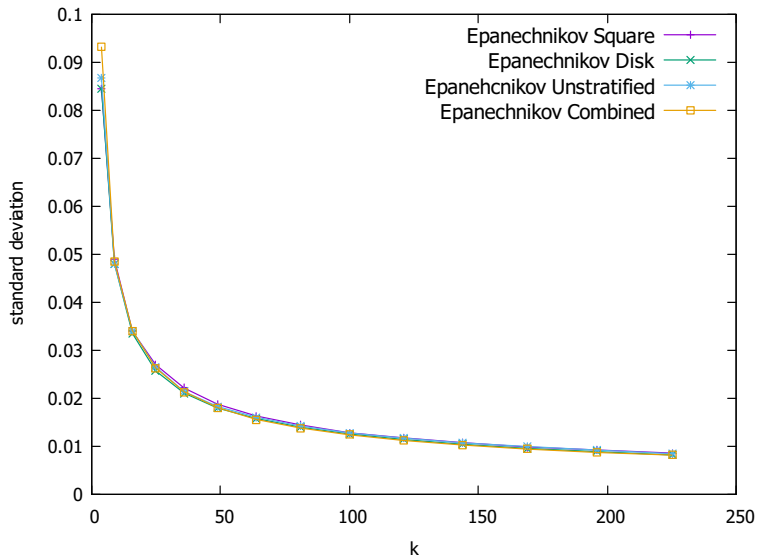


Figure 3.15.: Simulation of the Cornell Box for the Epanechnikov kernel.

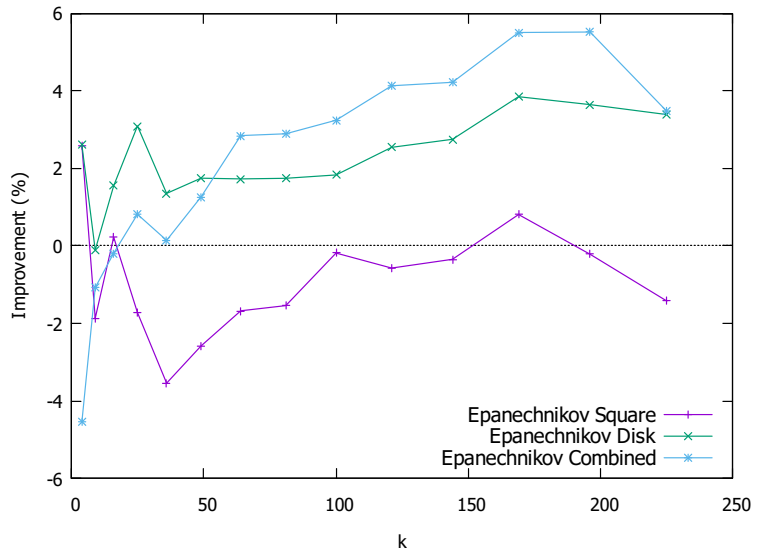


Figure 3.16.: Simulation of the Cornell Box for the Epanechnikov kernel in percentage.

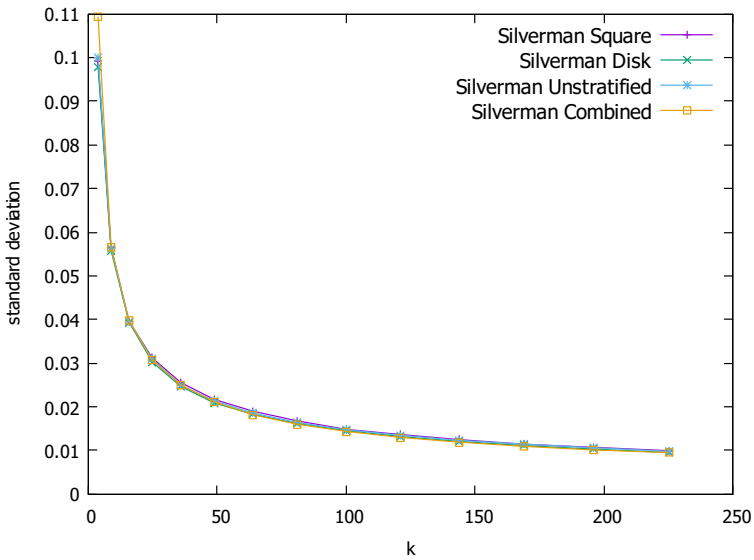


Figure 3.17.: Simulation of the Cornell Box for the Silverman kernel.

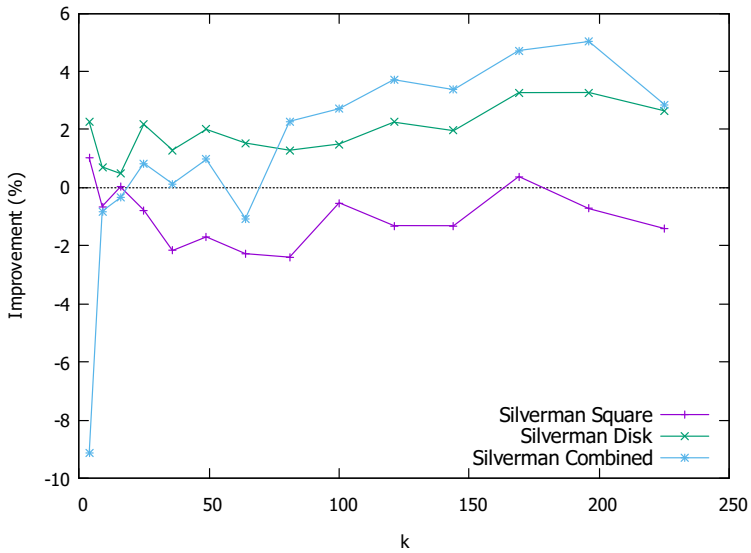


Figure 3.18.: Simulation of the Cornell Box for the Silverman kernel in percentage.

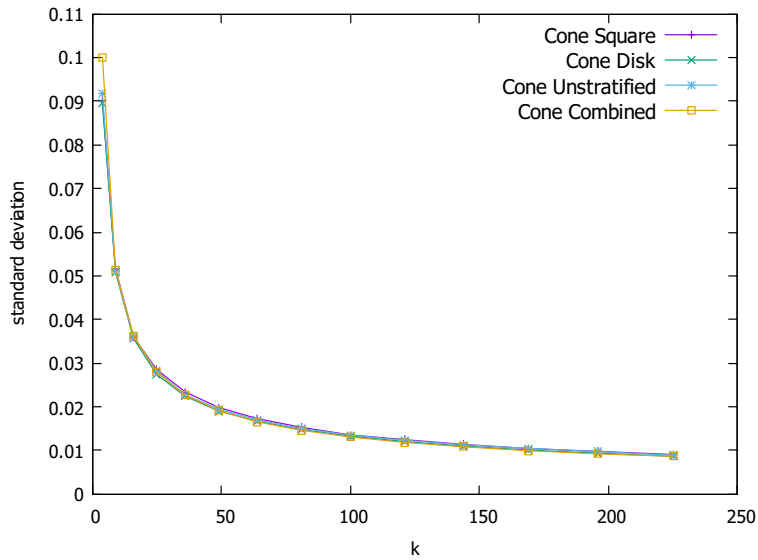


Figure 3.19.: Simulation of the Cornell Box for the Cone kernel.

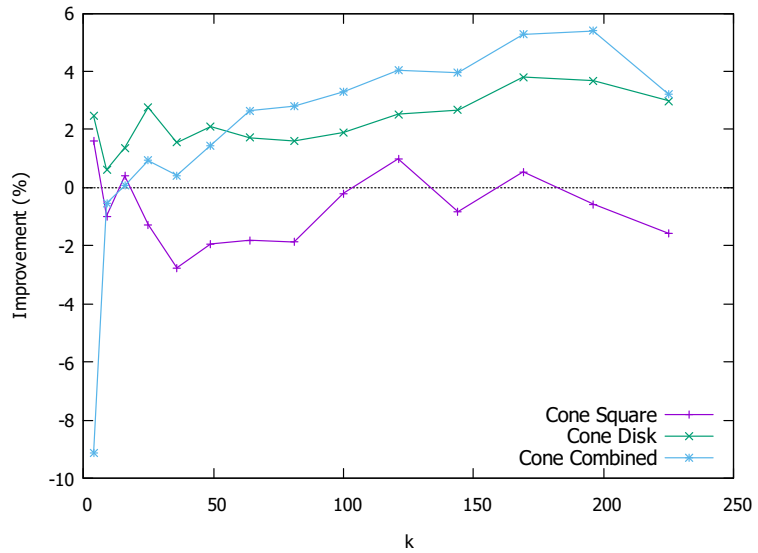


Figure 3.20.: Simulation of the Cornell Box for the Cone kernel in percentage.

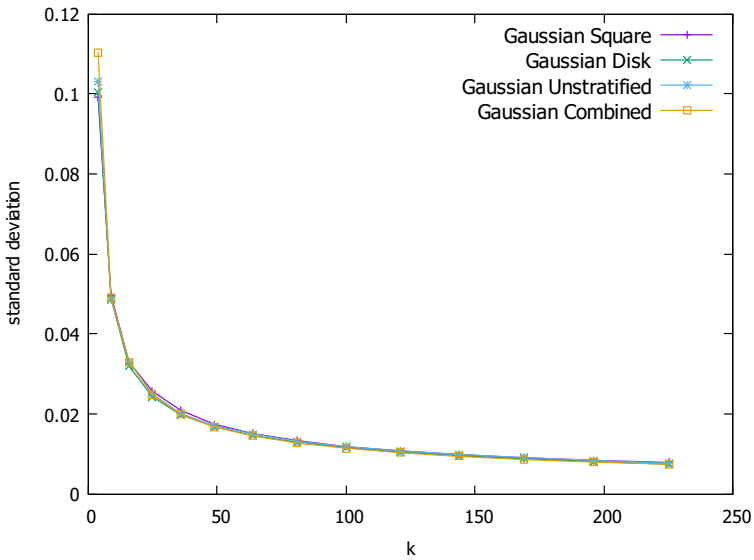


Figure 3.21.: Simulation of the Cornell Box for the Gaussian kernel.

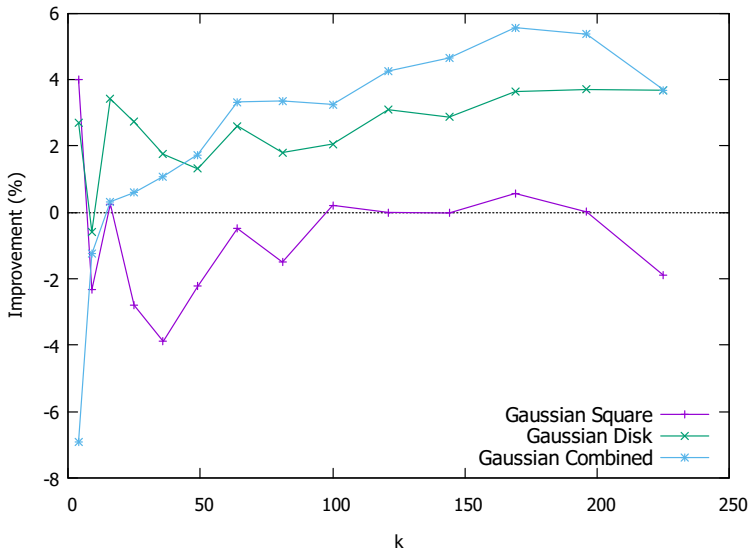


Figure 3.22.: Simulation of the Cornell Box for the Gaussian kernel in percentage.

Three hypotheses could explain this phenomenon, the first hypothesis is that stratified sampling reduces the overall variance and therefore the overall standard deviation of the scene, this means that there could be points for which an increase in the standard deviation could be observed, this taken into account that in this work the irradiance is only estimated for one point in the scene. The other hypothesis is that as in the empirical work on the over- and under-estimation biases this phenomenon could be just noise due to the small amount of iteration, only 10 000, whereas in the empirical work on the over- and under-estimation biases in some cases 150 000 000 iterations were needed to eliminate the noise (taking as an example the Silverman Kernel), however when encreasing the iterations from 10 000 to

100 000 and choosing $k = 225$ (a value of k for which the noise was lower) for the Gaussian kernel (taking into account that for this kernel 1 000 iterations were enough to reduce most of the noise in the empirical part of the over- and under-estimation biases) the standard deviation did not improve, proving wrong the second hypothesis. The third hypothesis is that there could be an unexpected interaction between the square stratification strategy and the kernels, but this doesn't seem to be the case either, since it is not clear that this stratification strategy has a positive impact on the standard deviation when running the code without a kernel as shown in Figure 3.23 (this means using a constant kernel).

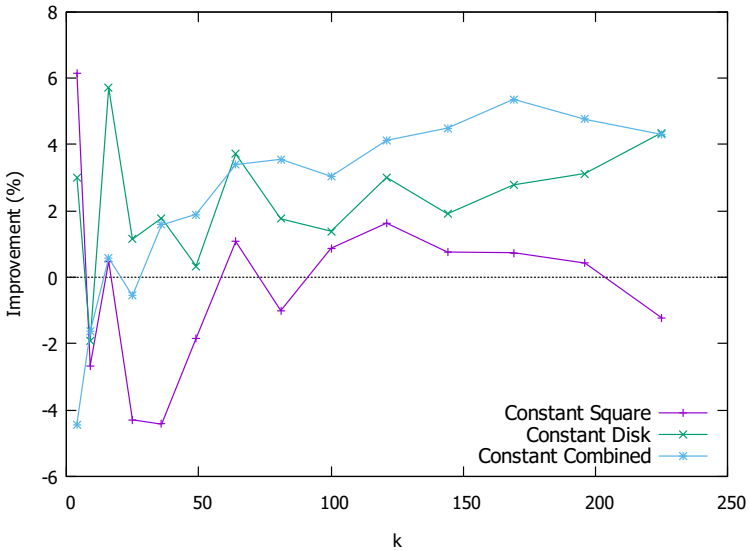


Figure 3.23.: Simulation of the Cornell Box for the constant kernel in percetage.

Now the three stratification strategies are analyzed separately and the kernels are compared to each other, also the behavior of the three stratification strategies is studied for the constant kernel.
For the square strategy, the Silverman kernel shows the highest standard deviation value followed by the Cone and Epanechnikov kernels, and the lowest value of standard deviation is obtained by the Gaussian and constant kernels. Notice that this follows the same order obtained in the first part of this empirical work and [Gar14].
The disk stratification strategy shows the same order observed in the square stratification strategy, the only changes are the magnitudes and percentages between the kernels. Also, an improvement in the standard deviation in all kernels is observed.
the greatest improvement of all strategies was obtained by the combined stratification, achieving in some cases more than 5% improvement in the standard deviation.

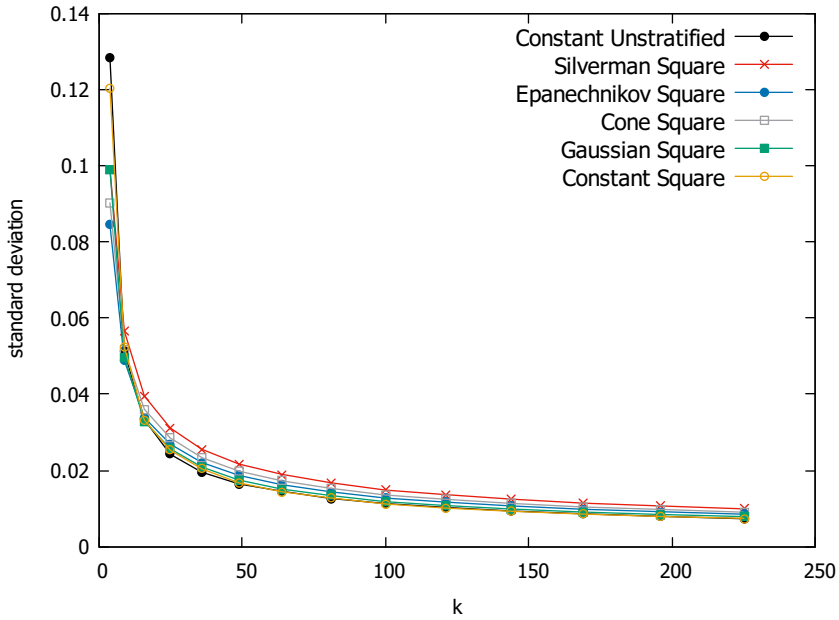


Figure 3.24.: Simulation of the Cornell Box for the square strategy.

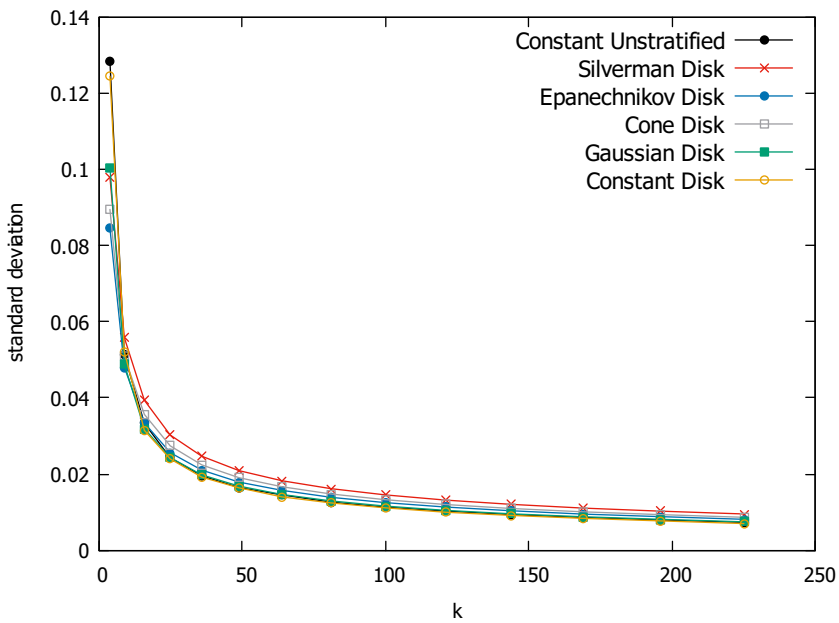


Figure 3.25.: Simulation of the Cornell Box for the disk strategy.

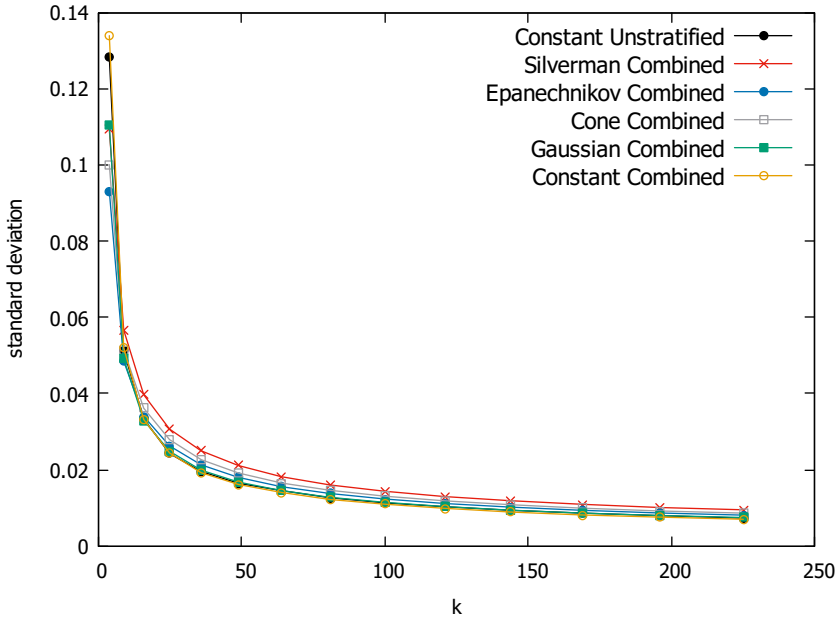


Figure 3.26.: Simulation of the Cornell Box for the combined strategy.

4. Conclusions and future work

It was theoretically and empirically proved that a bias in the estimation of the irradiance when using the 4 kernels studied in this work can not be in general eliminated for stratified sampling as opposed to random sampling, however a reduction of the bias was achieved when using kernels, for instance, the Silverman kernel reduced the bias to an extent that it was imperceptible. Although the source of the bias is known, a general solution to it remains unknown. until a solution is found the Silverman kernel can be used as workaround for the bias, since this kernel obtained the best results in random and stratified sampling. Also, it was empirically observed that the disk and combined stratification strategies improved the standard deviation of the irradiance estimate as the theory suggested, however, the square strategy had in most cases a negative impact on the standard deviation of the irradiance estimate in particular when using filtering kernels, the cause of this unexpected result remains unknown.

In future work a general solution for the over- and under-estimation biases will be studied, also the unexpected effects of the square stratification strategy on the standard deviation will be studied. It would also be interesting to study the effects of the disk stratification strategy in other global illumination algorithms.

A. Optimizations

Since the only difference between the 2D and 3D kernels is the normalization constant, only the optimization for the 2D kernels will be derived, understanding that a derivation of the 3D case can be obtained following the same steps.

A.1. Cone Kernel

$$\begin{aligned}
\sum_{i=1}^{k} \frac{\phi\left(1-\frac{r_i}{sr_k}\right)}{\pi\left(1-\frac{2}{3s}\right)r_k^2} &= \frac{\phi}{\pi\left(1-\frac{2}{3s}\right)}\left(\frac{\sum_{i=1}^{k} 1-\frac{r_i}{sr_k}}{r_k^2}\right) \\
&= \frac{\phi}{\pi\left(1-\frac{2}{3s}\right)}\left(\frac{\sum_{i=1}^{k} sr_k - r_i}{r_k^2\left(sr_k\right)}\right) \\
&= \frac{\phi}{\pi\left(1-\frac{2}{3s}\right)}\left(\frac{ksr_k - \sum_{i=1}^{k} r_i}{sr_k^3}\right) \\
&= \frac{\phi}{\pi\left(1-\frac{2}{3s}\right)}\left(\frac{-\sum_{i=1}^{k-1} r_i + ksr_k - r_k}{sr_k^3}\right) \\
&= \frac{\phi}{\pi\left(s-\frac{2}{3}\right)}\left[\frac{r_k(ks-1) - \sum_{i=1}^{k-1} r_i}{r_k^3}\right]
\end{aligned} \tag{A.1}$$

A.2. Gaussian Kernel

$$\begin{aligned}
\sum_{i=1}^{k} \frac{(\alpha\phi)\left(1-\frac{1-e^{-\frac{\beta r_i^2}{2r_k^2}}}{1-e^{-\beta}}\right)}{\pi r_k^2} &= \frac{\alpha\phi}{\pi}\left(\frac{\sum_{i=1}^{k}\left(1-\frac{1-e^{-\frac{\beta r_i^2}{2r_k^2}}}{1-e^{-\beta}}\right)}{r_k^2}\right) \\
&= \frac{\alpha\phi}{\pi}\left(\frac{\sum_{i=1}^{k}\frac{e^{-\frac{\beta r_i^2}{2r_k^2}}-e^{-\beta}}{1-e^{-\beta}}}{r_k^2}\right) \\
&= \frac{\alpha\phi}{\pi\left(1-e^{-\beta}\right)}\left[\frac{\sum_{i=1}^{k} e^{-\frac{\beta r_i^2}{2r_k^2}} - ke^{-\beta}}{r_k^2}\right]
\end{aligned} \tag{A.2}$$

A.3. Silverman Kernel

$$
\begin{aligned}
\sum_{i=1}^{k} \frac{3\phi\left(1-\left(\frac{r_i}{r_k}\right)^2\right)^2}{\pi r_k^2} &= \frac{3\phi}{\pi}\left(\frac{\sum_{i=1}^{k}\left(1-\left(\frac{r_i}{r_k}\right)^2\right)^2}{r_k^2}\right) \\
&= \frac{3\phi}{\pi}\left(\frac{\sum_{i=1}^{k}\frac{\left(r_k^2-r_i^2\right)^2}{r_k^4}}{r_k^2}\right) \\
&= \frac{3\phi}{\pi}\left(\frac{\sum_{i=1}^{k} r_k^4 - 2r_i^2 r_k^2 + r_i^4}{r_k^6}\right) \\
&= \frac{3\phi}{\pi}\left(\frac{-\sum_{i=1}^{k} 2r_i^2 r_k^2 + \sum_{i=1}^{k} r_i^4 + \mathrm{k}r_k^4}{r_k^6}\right) \\
&= \frac{3\phi}{\pi}\left[\frac{-2r_k^2\left(\sum_{i=1}^{k} r_i^2\right) + \sum_{i=1}^{k-1} r_i^4 + (k+1)r_k^4}{r_k^6}\right]
\end{aligned}
\tag{A.3}
$$

B. Mathematica code of the theoretical work

Stratified Photon Mapping:

Constant kernel:

```
(* 2D case*)
```
deriva

In[ ]:= $\int_{\sqrt{k-1}}^{\sqrt{k}} IP * 2 * r_k \, dr_k$

Out[ ]= $2\,IP\left(\frac{1-k}{2}+\frac{k}{2}\right)$

In[ ]:= $\text{Simplify}\left[2\,IP\left(\frac{1-k}{2}+\frac{k}{2}\right)\right]$
simplifica

Out[ ]= IP

In[ ]:= $\int_{\sqrt{k-1}}^{\sqrt{k}} \frac{k * IP}{r_k^2} * 2 * r_k \, dr_k$ (*radius $\sqrt{n}$*)

Out[ ]= IP k (-Log[-1 + k] + Log[k]) if condition

In[ ]:= $\int_{\frac{\sqrt{k-1}}{\sqrt{n}}}^{\frac{\sqrt{k}}{\sqrt{n}}} \frac{k * IP}{r_k^2 * n} * 2 * r_k * n \, dr_k$ (*radius 1*)

Out[ ]= IP k (-Log[-1 + k] + Log[k]) if condition

```
(*3D case*)
```
deriva

In[ ]:= $\int_{\sqrt[3]{k-1}}^{\sqrt[3]{k}} \frac{k * DP}{r_k^3} * 3 * r_k^2 \, dr_k$ (*radius $\sqrt{n}$*)

Out[ ]= DP k (-Log[-1 + k] + Log[k]) if condition

In[ ]:= $\int_{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}}^{\frac{\sqrt[3]{k}}{\sqrt[3]{n}}} \frac{k * DP}{r_k^3 * n} * 3 * r_k^2 * n \, dr_k$ (*radius 1*)

Out[ ]= DP k (-Log[-1 + k] + Log[k]) if condition

Epanechnikov kernel:

```
(*Definition of the Epanechnikov kernel for Photon Mapping*)
```
definición

$$K_{r_k}[r_] := \frac{2}{\pi * r_k^2} * \left(1 - (r_i * (1/r_k))^2\right)$$

```
(*Inner integral 2D case*)
```
producto interior ge··· deriva

In[•]:= $$\int_{\frac{\sqrt{i-1}}{\sqrt{n}}}^{\frac{\sqrt{i}}{\sqrt{n}}} \frac{2}{\pi * r_k^2} * \left(1 - \left(\frac{r_i}{r_k}\right)^2\right) * \phi * 2 * r_i * 2 * r_k * n^2 \, dr_i$$

Out[•]= $$\frac{2\phi}{\pi r_k^3} - \frac{4 i \phi}{\pi r_k^3} + \frac{4 n \phi}{\pi r_k}$$

In[•]:= $$\text{Simplify}\left[\frac{2\phi}{\pi r_k^3} - \frac{4 i \phi}{\pi r_k^3} + \frac{4 n \phi}{\pi r_k}\right]$$
simplifica

Out[•]= $$\frac{2\phi\left(1 - 2 i + 2 n r_k^2\right)}{\pi r_k^3}$$

```
(*Outer integral 2D case*)
```
producto exterior g··· deriva

In[•]:= $$\int_{\frac{\sqrt{k-1}}{\sqrt{n}}}^{\frac{\sqrt{k}}{\sqrt{n}}} \frac{2\phi\left(1 - 2 i + 2 n r_k^2\right)}{\pi r_k^3} \, dr_k$$

Out[•]= $$\frac{n\phi\,(1 - 2 i - 2(-1+k)\,k\,(\text{Log}[-1+k] - \text{Log}[k]))}{(-1+k)\,k\,\pi} \quad \text{if condition}$$

```
(*Sum up to the (k-1)th Photon*)
```
suma

In[•]:= $$\sum_{i=1}^{k-1} \frac{n\phi\,(1 - 2 i - 2(-1+k)\,k\,(\text{Log}[-1+k] - \text{Log}[k]))}{(-1+k)\,k\,\pi} \quad \text{(*result with sum method*)}$$

Out[•]= $$-\frac{(-1+k)\,n\phi\,(1 + 2 k\,\text{Log}[-1+k] - 2 k\,\text{Log}[k])}{k\,\pi}$$

```
(*Inner integral over the whole interval 2D case*)
```
producto interior generalizado deriva

In[•]:= $$\int_{0}^{\frac{\sqrt{k-1}}{\sqrt{n}}} \frac{2}{\pi * r_k^2} * \left(1 - \left(\frac{r_i}{r_k}\right)^2\right) * \phi * 2 * r_i * 2 * r_k * n^2 \, dr_i$$

Out[•]= $$-\frac{2(-1+k)\,\phi\left(-1 + k - 2 n r_k^2\right)}{\pi r_k^3}$$

```
(*Outer integral based on the whole interval 2D case*)
```
producto exterior generalizado deriva

In[ ]:= $\int_{\frac{\sqrt{k-1}}{\sqrt{n}}}^{\frac{\sqrt{k}}{\sqrt{n}}} -\frac{2\,(-1+k)\,\phi\,\left(-1+k-2\,n\,r_k^2\right)}{\pi\,r_k^3}\,dr_k$ (*result with integral method*)

Out[ ]= $-\frac{(-1+k)\,\phi\,\left(n+4\,k\,n\,\mathrm{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]-4\,k\,n\,\mathrm{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)}{k\,\pi}$ if condition

(*3D*)
deriv

(*Inner Integral 3D case*)
produ··· integral deriva

In[ ]:= $\int_0^{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}} \frac{15}{8*\pi*r_k^3}*\left(1-\left(\frac{r_i}{r_k}\right)^2\right)*\phi*3*r_i^2*3*r_k^2*n^2\,dr_i$

Out[ ]= $\frac{9\,(-1+k)\,n^{1/3}\,\phi\,\left(-3\,(-1+k)^{2/3}+5\,n^{2/3}\,r_k^2\right)}{8\,\pi\,r_k^3}$

(*Outer integral 3D case*)
producto exterior g··· deriva

In[ ]:= $\int_{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}}^{\frac{\sqrt[3]{k}}{\sqrt[3]{n}}} \left(\frac{9\,(-1+k)\,n^{1/3}\,\phi\,\left(-3\,(-1+k)^{2/3}+5\,n^{2/3}\,r_k^2\right)}{8\,\pi\,r_k^3}\right)dr_k$

Out[ ]= $\frac{3\,(-1+k)\,n\,\phi\,\left(9\,(-1+k)^{2/3}+k^{2/3}\,(-9-10\,\mathrm{Log}[-1+k]+10\,\mathrm{Log}[k])\right)}{16\,k^{2/3}\,\pi}$ if condition

(*Inner integral for the normalization costant 2D case*)
producto interior generalizado deriva

In[ ]:= $\int_0^{r_k}\left(1-\left(\frac{r_i}{r_k}\right)^2\right)r_i\,dr_i$

Out[ ]= $\frac{r_k^2}{4}$

(*Outer integral for the normalization constant 2D case*)
producto exterior generalizado deriva

In[ ]:= $\int_0^{2\pi}\left(\frac{r_k^2}{4}\right)d\theta$

$\frac{\pi\,r_k^2}{2}$ (*2D Normalization*)
deriva

(*Inner integral for the normalization constant 3D case*)
producto interior generalizado deriva

In[∘]:= $\int_0^{r_k} \left(1 - \left(\frac{r_i}{r_k}\right)^2\right) r_i^2 * \text{Sin}[\phi]\, dr_i$
seno

Out[∘]= $\frac{2}{15} \text{Sin}[\phi]\, r_k^3$

```
(*Second inner integral for the normalization constant 3D case*)
```
deriva

In[∘]:= $\int_0^{\pi} \left(\frac{2}{15} \text{Sin}[\phi]\, r_k^3\right) d\phi$
seno

Out[∘]= $\frac{4\, r_k^3}{15}$

```
(*Outer integral for the nomalization constant 3D case*)
```
producto exterior generalizado deriva

In[∘]:= $\int_0^{2\pi} \left(\frac{4\, r_k^3}{15}\right) d\theta$

$\frac{8\, \pi\, r_k^3}{15}$ (*3D Normalization*)
deriva

Cone filter:

```
(*Definition of the Cone filter for Photon Mapping*)
```
definición cono

In[∘]:= $K_{r_k}[d_] := \frac{1 - \frac{d}{s * r_k}}{\pi * r_k^2 * \left(1 - \frac{2}{3 * s}\right)}$

```
(*Inner integral 2D case*)
```
producto interior ge··· deriva

$\int_0^{\frac{\sqrt{k-1}}{\sqrt{n}}} \frac{1 - \frac{r_i}{s * r_k}}{\pi * r_k^2 * \left(1 - \frac{2}{3 * s}\right)} * \phi * 2 * r_i * 2 * r_k * n^2\, dr_i$

Out[∘]= $\frac{2\,(-1+k)\,\sqrt{n}\,\phi\,\left(-2\,\sqrt{-1+k} + 3\,\sqrt{n}\, s\, r_k\right)}{\pi\,(-2+3\,s)\, r_k^2}$

In[∘]:= FullSimplify$\left[\frac{2\,(-1+k)\,\sqrt{n}\,\phi\,\left(-2\,\sqrt{-1+k} + 3\,\sqrt{n}\, s\, r_k\right)}{\pi\,(-2+3\,s)\, r_k^2}\right]$
simplifica completamente

Out[∘]= $\frac{2\,(-1+k)\,\sqrt{n}\,\phi\,\left(-2\,\sqrt{-1+k} + 3\,\sqrt{n}\, s\, r_k\right)}{\pi\,(-2+3\,s)\, r_k^2}$

```
(*Outer integral 2D case*)
```
producto exterior g··· deriva

$$\int_{\frac{\sqrt{k-1}}{\sqrt{n}}}^{\frac{\sqrt{k}}{\sqrt{n}}} \frac{2\,(-1+k)\,\sqrt{n}\,\phi\left(-2\sqrt{-1+k}+3\sqrt{n}\,s\,r_k\right)}{\pi\,(-2+3\,s)\,r_k^2}\,dr_k \quad \text{(*result with integral method*)}$$

Out[•]= $$\frac{(-1+k)\,n\,\phi\left(-4+4\sqrt{\frac{-1+k}{k}}-3\,s\,\mathrm{Log}[-1+k]+3\,s\,\mathrm{Log}[k]\right)}{\pi\,(-2+3\,s)} \text{ if } \boxed{\text{condition}}$$

```
(*Integral for the k th photon 2D case*)
```
integral deriva

In[•]:= $$\int_{\frac{\sqrt{k-1}}{\sqrt{n}}}^{\frac{\sqrt{k}}{\sqrt{n}}} \frac{1-\frac{r_k}{s*r_k}}{\pi*r_k^2*\left(1-\frac{2}{3*s}\right)}*\phi*2*r_k*n\,dr_k$$

Out[•]= $$\frac{6\,n\,(-1+s)\,\phi\left(-\mathrm{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+\mathrm{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)}{\pi\,(-2+3\,s)} \text{ if } \boxed{\text{condition}}$$

```
(*Sum of the result of the photons up to
```
suma

```
 the (k-1) th  and the result of the k th photon*)
```

In[•]:= $$\frac{6\,n\,(-1+s)\,\phi\left(-\mathrm{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+\mathrm{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)}{\pi\,(-2+3\,s)}+\frac{(-1+k)\,n\,\phi\left(-4+4\sqrt{\frac{-1+k}{k}}-3\,s\,\mathrm{Log}[-1+k]+3\,s\,\mathrm{Log}[k]\right)}{\pi\,(-2+3\,s)}$$

Out[•]= $$\frac{(-1+k)\,n\,\phi\left(-4+4\sqrt{\frac{-1+k}{k}}-3\,s\,\mathrm{Log}[-1+k]+3\,s\,\mathrm{Log}[k]\right)}{\pi\,(-2+3\,s)}+\frac{6\,n\,(-1+s)\,\phi\left(-\mathrm{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+\mathrm{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)}{\pi\,(-2+3\,s)}$$

In[•]:= $$\mathrm{FullSimplify}\left[\frac{(-1+k)\,n\,\phi\left(-4+4\sqrt{\frac{-1+k}{k}}-3\,s\,\mathrm{Log}[-1+k]+3\,s\,\mathrm{Log}[k]\right)}{\pi\,(-2+3\,s)}+\frac{6\,n\,(-1+s)\,\phi\left(-\mathrm{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+\mathrm{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)}{\pi\,(-2+3\,s)}\right]$$

simplifica completamente

Out[•]= $$\frac{n\,\phi\left(\frac{4\left(\sqrt{-1+k}-\sqrt{k}\right)(-1+k)}{\sqrt{k}}+(3-3\,k\,s)\,\mathrm{Log}[-1+k]+3\,(-1+k\,s)\,\mathrm{Log}[k]\right)}{\pi\,(-2+3\,s)}$$

```
(*Inner integral 3D case*)
```
producto interior ge··· deriva

In[∘]:=
$$\int_0^{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}} \frac{\left(1-\frac{r_i}{s*r_k}\right)}{\frac{\pi\,(-3+4\,s)\,r_k^3}{3\,s}} * \phi * 3 * r_i^2 * 3 * r_k^2 * n^2\, dr_i \quad (* \text{ 3D Case } *)$$

deriva

Out[∘]=
$$\frac{9\,(-1+k)\,n^{2/3}\,\phi\left(-3\,(-1+k)^{1/3}+4\,n^{1/3}\,s\,r_k\right)}{4\,\pi\,(-3+4\,s)\,r_k^2}$$

(*Outer integral 3D case*)

producto exterior g··· deriva

In[∘]:=
$$\int_{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}}^{\frac{\sqrt[3]{k}}{\sqrt[3]{n}}} \left(\frac{9\,(-1+k)\,n^{2/3}\,\phi\left(-3\,(-1+k)^{1/3}+4\,n^{1/3}\,s\,r_k\right)}{4\,\pi\,(-3+4\,s)\,r_k^2}\right) dr_k$$

Out[∘]=
$$\frac{3\,(-1+k)\,n\,\phi\left(-9+9\left(\frac{-1+k}{k}\right)^{1/3}-4\,s\,\text{Log}[-1+k]+4\,s\,\text{Log}[k]\right)}{4\,\pi\,(-3+4\,s)} \text{ if } \boxed{\text{condition}}$$

(*Integral for the k th photon 3D case*)

integral deriva

In[∘]:=
$$\int_{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}}^{\frac{\sqrt[3]{k}}{\sqrt[3]{n}}} \frac{\left(1-\frac{1}{s}\right)}{\frac{\pi\,(-3+4\,s)\,r_k^3}{3\,s}} * \phi * 3 * r_k^2 * n\, dr_k$$

Out[∘]=
$$\frac{3\,n\,(-1+s)\,\phi\,(-\text{Log}[-1+k]+\text{Log}[k])}{\pi\,(-3+4\,s)} \text{ if } \boxed{\text{condition}}$$

(*Sum of the result of the photons up to the (k-1)th and the k th photon*)

suma

In[∘]:=
$$\frac{3\,(-1+k)\,n\,\phi\left(-9+9\left(\frac{-1+k}{k}\right)^{1/3}-4\,s\,\text{Log}[-1+k]+4\,s\,\text{Log}[k]\right)}{4\,\pi\,(-3+4\,s)} + \frac{3\,n\,(-1+s)\,\phi\,(-\text{Log}[-1+k]+\text{Log}[k])}{\pi\,(-3+4\,s)}$$

Out[∘]=
$$\frac{3\,n\,(-1+s)\,\phi\,(-\text{Log}[-1+k]+\text{Log}[k])}{\pi\,(-3+4\,s)} + \frac{3\,(-1+k)\,n\,\phi\left(-9+9\left(\frac{-1+k}{k}\right)^{1/3}-4\,s\,\text{Log}[-1+k]+4\,s\,\text{Log}[k]\right)}{4\,\pi\,(-3+4\,s)}$$

In[ ]:= FullSimplify[$\frac{3\,n\,(-1+s)\,\phi\,(-\text{Log}[-1+k]+\text{Log}[k])}{\pi\,(-3+4\,s)}$ +

$\frac{3\,(-1+k)\,n\,\phi\left(-9+9\left(\frac{-1+k}{k}\right)^{1/3}-4\,s\,\text{Log}[-1+k]+4\,s\,\text{Log}[k]\right)}{4\,\pi\,(-3+4\,s)}$]

simplifica completamente

Out[ ]= $\frac{1}{4\,\pi\,(-3+4\,s)}\,3\,n\,\phi\left(9\left(-1+\left(\frac{-1+k}{k}\right)^{1/3}\right)(-1+k)+(4-4\,k\,s)\,\text{Log}[-1+k]+4\,(-1+k\,s)\,\text{Log}[k]\right)$

(*Inner integral for the normalization constant 2D case*)

producto interior generalizado deriva

In[ ]:= $\int_0^{r_k}\left(1-\frac{r_i}{s*r_k}\right)r_i\,dr_i$

Out[ ]= $\frac{r_k^2}{2}-\frac{r_k^2}{3\,s}$

(*Outer integral for the normalization constant 2D case*)

producto exterior generalizado deriva

In[ ]:= $\int_0^{2\pi}\left(\frac{r_k^2}{2}-\frac{r_k^2}{3\,s}\right)d\theta$

$2\,\pi\left(\frac{r_k^2}{2}-\frac{r_k^2}{3\,s}\right)$ (* 2D Normalization *)

deriva

(*Inner integral for the normalization constant 3D case*)

producto interior generalizado deriva

In[ ]:= $\int_0^{r_k}\left(1-\frac{r_i}{s*r_k}\right)r_i^2*\text{Sin}[\phi]\,dr_i$

seno

Out[ ]= $\frac{1}{3}\,\text{Sin}[\phi]\,r_k^3-\frac{\text{Sin}[\phi]\,r_k^3}{4\,s}$

(*Sencond inner integral for the normalization constant 3D case*)

deriva

In[ ]:= $\int_0^{\pi}\left(\frac{1}{3}\,\text{Sin}[\phi]\,r_k^3-\frac{\text{Sin}[\phi]\,r_k^3}{4\,s}\right)d\phi$

seno

Out[ ]= $\frac{(-3+4\,s)\,r_k^3}{6\,s}$

(*Outer integral for the normalization constant 3D case*)

producto exterior generalizado deriva

In[∘]:= $\int_0^{2\pi} \left(\frac{(-3+4s)\, r_k^3}{6s} \right) d\theta$

$\frac{\pi\,(-3+4s)\, r_k^3}{3s}$ (* 3D Normalization *)
deriva

Silverman kernel:

(*Definition of the Silverman kernel for Photon Mapping*)
definición

In[∘]:= $K_{r_k}[d_] := \frac{3}{\pi * r_k^2} * \left(1 - \left(\frac{d}{r_k}\right)^2\right)^2$

(*Inner integral 2D case*)
producto interior ge··· deriva

$$\int_{\frac{\sqrt{i-1}}{\sqrt{n}}}^{\frac{\sqrt{i}}{\sqrt{n}}} \frac{3}{\pi * r_k^2} * \left(1 - \left(\frac{r_i}{r_k}\right)^2\right)^2 * \phi * 2 * r_i * 2 * r_k * n^2\, dr_i$$

Out[∘]= $\frac{2\phi}{n\pi r_k^5} - \frac{6 i \phi}{n\pi r_k^5} + \frac{6 i^2 \phi}{n\pi r_k^5} + \frac{6\phi}{\pi r_k^3} - \frac{12 i \phi}{\pi r_k^3} + \frac{6 n \phi}{\pi r_k}$

In[∘]:= FullSimplify$\left[\frac{2\phi}{n\pi r_k^5} - \frac{6 i \phi}{n\pi r_k^5} + \frac{6 i^2 \phi}{n\pi r_k^5} + \frac{6\phi}{\pi r_k^3} - \frac{12 i \phi}{\pi r_k^3} + \frac{6 n \phi}{\pi r_k}\right]$
simplifica completamente

Out[∘]= $\frac{2\phi\left(1 + 3(-1+i)\, i + 3 n r_k^2 \left(1 - 2i + n r_k^2\right)\right)}{n\pi r_k^5}$

(*Outer integral 2D case*)
producto exterior g··· deriva

In[∘]:= $\int_{\frac{\sqrt{k-1}}{\sqrt{n}}}^{\frac{\sqrt{k}}{\sqrt{n}}} \frac{2\phi\left(1 + 3(-1+i)\, i + 3 n r_k^2 \left(1 - 2i + n r_k^2\right)\right)}{n\pi r_k^5}\, dr_k$

Out[∘]= $\frac{1}{2(-1+k)^2 k^2 \pi}$
$n\phi\left(-1 - 3(-1+i)\, i - 4k + 6 i (1+i)\, k + 6(1-2i)\, k^2 - 6(-1+k)^2 k^2 (\mathrm{Log}[-1+k] - \mathrm{Log}[k])\right)$
if condition

(*Sum up to the (k-1)th photon*)
suma

In[∘]:= $\sum_{i=1}^{k-1} \frac{1}{2(-1+k)^2 k^2 \pi}$
$n\phi\left(-1 - 3(-1+i)\, i - 4k + 6 i (1+i)\, k + 6(1-2i)\, k^2 - 6(-1+k)^2 k^2 (\mathrm{Log}[-1+k] - \mathrm{Log}[k])\right)$
logaritmo logaritmo

$$-\frac{(-1+k)\, n\, \phi \left(1+4k+6k^2 \operatorname{Log}[-1+k]-6k^2 \operatorname{Log}[k]\right)}{2k^2 \pi}$$ (*reuslt with the sum method*)

(*Inner integral for the whole interval 2D case*)

producto interior generalizado deriva

In[ ]:=
$$\int_0^{\frac{\sqrt{k-1}}{\sqrt{n}}} \frac{3}{\pi * r_k{}^2} * \left(1-\left(\frac{r_i}{r_k}\right)^2\right)^2 * \phi * 2 * r_i * 2 * r_k * n^2 \, dr_i$$

Out[ ]=
$$\frac{2(-1+k)\,\phi\left((-1+k)^2-3(-1+k)\, n\, r_k^2+3n^2 r_k^4\right)}{n\,\pi\, r_k^5}$$

(*Outer integral based on the whole interval 2D case*)

producto exterior generalizado deriva

$$\int_{\frac{\sqrt{k-1}}{\sqrt{n}}}^{\frac{\sqrt{k}}{\sqrt{n}}} \frac{2(-1+k)\,\phi\left((-1+k)^2-3(-1+k)\, n\, r_k^2+3n^2 r_k^4\right)}{n\,\pi\, r_k^5}\, dr_k$$ (*result with integral method*)

Out[ ]=
$$\frac{(-1+k)\,\phi\left(-n^2-4kn^2-12k^2n^2 \operatorname{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+12k^2n^2 \operatorname{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)}{2k^2 n\,\pi}$$ if condition

(*Inner integral 3D case*)

producto interior ge··· deriva

$$\int_0^{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}} \frac{105}{32\, r_k^3} * \left(1-\left(\frac{r_i}{r_k}\right)^2\right)^2 * \phi * 3 * r_i{}^2 * 3 * r_k{}^2 * n^2 \, dr_i$$ (*3D Case*)

deriva

Out[ ]=
$$\frac{9(-1+k)\,\phi\left(15(-1+k)^{4/3}-42(-1+k)^{2/3} n^{2/3} r_k^2+35 n^{4/3} r_k^4\right)}{32 n^{1/3} r_k^5}$$

(*Outer integral 3D case*)

producto exterior g··· deriva

In[ ]:=
$$\int_{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}}^{\frac{\sqrt[3]{k}}{\sqrt[3]{n}}} \left(\frac{9(-1+k)\,\phi\left(15(-1+k)^{4/3}-42(-1+k)^{2/3} n^{2/3} r_k^2+35 n^{4/3} r_k^4\right)}{32 n^{1/3} r_k^5}\right) dr_k$$

Out[ ]=
$$\frac{1}{32 n^{1/3}} 9(-1+k)\,\phi\left(-\frac{1}{4} n^{4/3}\left(69+140 \operatorname{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]\right)+\frac{1}{4} n^{4/3}\left(\frac{3\left(5+28(-1+k)^{1/3} k^{2/3}-5k\right)(-1+k)^{1/3}}{k^{4/3}}+140 \operatorname{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\right)$$ if condition

(*Inner integral for the normalization constant 2D case*)

producto interior generalizado deriva

In[ ]:= $\int_0^{r_k} \left(\left(1 - \left(\frac{r_i}{r_k} \right)^2 \right)^2 * r_i \right) dr_i$

Out[ ]= $\frac{r_k^2}{6}$

```
(*Outer integral for the normalization constant 2D case*)
```
producto exterior generalizado deriva

In[ ]:= $\int_0^{2\pi} \frac{r_k^2}{6} d\theta$

$\frac{\pi r_k^2}{3}$ (*2D Normalization*)

deriva

```
(*Inner integral for the normalization cosntant 3D case*)
```
producto interior generalizado deriva

In[ ]:= $\int_0^{r_k} \left(\left(1 - \left(\frac{r_i}{r_k} \right)^2 \right)^2 * r_i^2 * \mathrm{Sin}[\phi] \right) dr_i$

seno

Out[ ]= $\frac{8}{105} \mathrm{Sin}[\phi]\, r_k^3$

```
(*Second inner integral for  the normalization constant 3D case*)
```
deriva

In[ ]:= $\int_0^{\pi} \frac{8}{105} \mathrm{Sin}[\phi]\, r_k^3\, d\phi$

seno

Out[ ]= $\frac{16 r_k^3}{105}$

```
(*Outer integral for  the normalization constant 3D case*)
```
producto exterior generalizado deriva

In[ ]:= $\int_0^{2\pi} \frac{16 r_k^3}{105} d\theta$

$\frac{32 \pi r_k^3}{105}$ (*3D Normalization*)

deriva

Gaussian filter:

```
(*Normalization constat 2D case*)
```
deriva

In[ ]:= $\alpha = 1.728$

Out[ ]= 1.728

```
(*Constant used in the  Gaussian filter*)
```
constante

In[∘]:= $\beta = 1.95$

Out[∘]= 1.95

```
(*Normalization constant 3D case*)
```
deriva

In[∘]:= $$\alpha_v = \frac{2\left(-1+E^{\beta}\right)\beta^{3/2}}{\left(-2\sqrt{\beta}\left(3E^{\beta/2}+\beta\right)+3E^{\beta}\sqrt{2\pi}\,\mathrm{Erf}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]\right)}$$

Out[∘]= 1.97758

```
(*Definition of the Gaussian filter for Photon Mapping*)
```
definición

In[∘]:= $$K_{r_k}[d_] := \frac{1}{\pi * r_k^2} * \alpha * \left(1 - \left(\frac{1 - E^{-\beta * d^2 * \frac{1}{2 * r_k^2}}}{1 - E^{-\beta}}\right)\right)$$

```
(*Inner inegral 2D case*)
```
producto interior g··· deriva

In[∘]:= $$\int_0^{\frac{\sqrt{k-1}}{\sqrt{n}}} \frac{1}{\pi * r_k^2} * \alpha * \left(1 - \left(\frac{1 - E^{-\beta * r_i^2 * \frac{1}{2 * r_k^2}}}{1 - E^{-\beta}}\right)\right) * \phi * 2 * r_i * 2 * r_k * n^2 \, dr_i$$

Out[∘]= $$-\frac{2 n^2 \alpha \phi \left(\frac{(-1+k)\beta}{n} + 2E^{\beta}\left(-1 + E^{-\frac{(-1+k)\beta}{2 n r_k^2}}\right) r_k^2\right)}{\left(-1+E^{\beta}\right)\pi \beta r_k}$$

In[∘]:= $$\text{FullSimplify}\left[-\frac{2 n^2 \alpha \phi \left(\frac{(-1+k)\beta}{n} + 2E^{\beta}\left(-1 + E^{-\frac{(-1+k)\beta}{2 n r_k^2}}\right) r_k^2\right)}{\left(-1+E^{\beta}\right)\pi \beta r_k}\right]$$
simplifica completamente

Out[∘]= $$-\frac{2 n^2 \alpha \phi \left(\frac{(-1+k)\beta}{n} + 2E^{\beta}\left(-1 + E^{-\frac{(-1+k)\beta}{2 n r_k^2}}\right) r_k^2\right)}{\left(-1+E^{\beta}\right)\pi \beta r_k}$$

```
(*Outer integral 2D case*)
```
producto exterior g··· deriva

$$\int_{\frac{\sqrt{k-1}}{\sqrt{n}}}^{\frac{\sqrt{k}}{\sqrt{n}}} -\frac{2\,n^2\,\alpha\,\phi\left(\frac{(-1+k)\,\beta}{n}+2\,E^{\beta}\left(-1+E^{-\frac{(-1+k)\,\beta}{2\,n\,r_k^2}}\right)r_k^2\right)}{\left(-1+E^{\beta}\right)\pi\,\beta\,r_k}\,dr_k$$ (*result with integral mehtod*)

Out[ ]=
$$-\frac{1}{\left(-1+E^{\beta}\right)\pi\,\beta}\,n\,\alpha\,\phi\left(E^{\beta}\,(-1+k)\,\beta\,\text{Gamma}\left[0,\frac{\beta}{2}\right]-E^{\beta}\,(-1+k)\,\beta\,\text{Gamma}\left[0,\frac{(-1+k)\,\beta}{2\,k}\right]-2\left(E^{\beta/2}\left(-1+E^{\beta/2}+k-E^{\frac{\beta}{2\,k}}\,k\right)+(-1+k)\,\beta\,\text{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+(\beta-k\,\beta)\,\text{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)\right)$$ if condition

$$-n\,\alpha\,\phi\left(\left(E^{\beta}\,(-1+k)\,\beta\,\text{Gamma}\left[0,\frac{\beta}{2}\right]\right)\Big/\left(-1+E^{\beta}\right)\pi\,\beta-\left(E^{\beta}\,(-1+k)\,\beta\,\text{Gamma}\left[0,\frac{(-1+k)\,\beta}{2\,k}\right]\right)\Big/\left(-1+E^{\beta}\right)\pi\,\beta-2\left(E^{\beta/2}\left(-1+E^{\beta/2}+k-E^{\frac{\beta}{2\,k}}\,k\right)\Big/\left(-1+E^{\beta}\right)\pi\,\beta+(-1+k)\,\beta\,\text{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]\Big/\left(-1+E^{\beta}\right)\pi\,\beta+(\beta-k\,\beta)\,\text{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\Big/\left(-1+E^{\beta}\right)\pi\,\beta\right)\right)$$

gamma de Euler

gamma de Euler

logaritmo

logaritmo

(*Integral for the k th photon*)

integral

In[ ]:=
$$\int_{\frac{\sqrt{k-1}}{\sqrt{n}}}^{\frac{\sqrt{k}}{\sqrt{n}}} \frac{1}{\pi * r_k^{\,2}} * \alpha * \left(1-\left(\frac{1-E^{-\beta*\frac{1}{2}}}{1-E^{-\beta}}\right)\right) * \phi * 2 * r_k * n\,dr_k$$

Out[ ]=
$$\frac{2\,n\,\alpha\,\phi\left(-\text{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+\text{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)}{\pi+E^{\beta/2}\,\pi}$$ if condition

(*Summing the result up to the k th photon*)

In[10]:=
$$\frac{2\,n\,\alpha\,\phi\left(-\text{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+\text{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)}{\pi+E^{\beta/2}\,\pi}+$$
$$-\frac{1}{\left(-1+E^{\beta}\right)\pi\,\beta}\,n\,\alpha\,\phi\left(E^{\beta}\,(-1+k)\,\beta\,\text{Gamma}\left[0,\frac{\beta}{2}\right]-E^{\beta}\,(-1+k)\,\beta\,\text{Gamma}\left[0,\frac{(-1+k)\,\beta}{2\,k}\right]-\right.$$
gamma de Euler gamma de Euler
$$\left.2\left(E^{\beta/2}\left(-1+E^{\beta/2}+k-E^{\frac{\beta}{2k}}\,k\right)+(-1+k)\,\beta\,\text{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+(\beta-k\,\beta)\,\text{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)\right)$$
logaritmo logaritmo

Out[10]=
$$\frac{2\,n\,\alpha\,\phi\left(-\text{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+\text{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)}{\pi+e^{\beta/2}\,\pi}-\frac{1}{\left(-1+e^{\beta}\right)\pi\,\beta}$$
$$n\,\alpha\,\phi\left(e^{\beta}\,(-1+k)\,\beta\,\text{Gamma}\left[0,\frac{\beta}{2}\right]-e^{\beta}\,(-1+k)\,\beta\,\text{Gamma}\left[0,\frac{(-1+k)\,\beta}{2\,k}\right]-\right.$$
$$\left.2\left(e^{\beta/2}\left(-1+e^{\beta/2}+k-e^{\frac{\beta}{2k}}\,k\right)+(-1+k)\,\beta\,\text{Log}\left[\frac{\sqrt{-1+k}}{\sqrt{n}}\right]+(\beta-k\,\beta)\,\text{Log}\left[\frac{\sqrt{k}}{\sqrt{n}}\right]\right)\right)$$

In[11]:= `FullSimplify[%10]`
simplifica completamente

Out[11]=
$$-\frac{1}{\left(-1+e^{\beta}\right)\pi\,\beta}\,n\,\alpha\,\phi\left(-2\,e^{\beta/2}\left(-1+e^{\beta/2}+k-e^{\frac{\beta}{2k}}\,k\right)+e^{\beta}\,(-1+k)\,\beta\,\text{ExpIntegralE}\left[1,\frac{\beta}{2}\right]-\right.$$
$$\left.e^{\beta}\,(-1+k)\,\beta\,\text{ExpIntegralE}\left[1,\frac{(-1+k)\,\beta}{2\,k}\right]+\left(e^{\beta/2}-k\right)\beta\,(\text{Log}[-1+k]-\text{Log}[k])\right)$$

```
(*Inner integral 3D case*)
```
producto interior ge··· deriva

In[ ]:=
$$\int_{0}^{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}}\left(\frac{3}{4*\pi*r_k^{3}}*\alpha_v*\left(1-\left(\frac{1-E^{-\beta*r_i^{2}*\frac{1}{2*r_k^{2}}}}{1-E^{-\beta}}\right)\right)*\phi*3*r_i^{2}*3*r_k^{2}*n^{2}\right)dr_i$$

Out[ ]=
$$\frac{1}{8\left(-1+E^{\beta}\right)\pi\,r_k}\,9\,n\,\phi\left(2-2\,k-\frac{6\,E^{\beta-\frac{(-1+k)^{2/3}\,\beta}{2\,n^{2/3}\,r_k^{2}}}\,(-1+k)^{1/3}\,n^{2/3}\,r_k^{2}}{\beta}+\frac{3\,E^{\beta}\,n\,\sqrt{2\,\pi}\,\text{Erf}\left[\frac{(-1+k)^{1/3}\,\sqrt{\beta}}{\sqrt{2}\,n^{1/3}\,r_k}\right]r_k^{3}}{\beta^{3/2}}\right)\alpha_v$$

```
(*Outer integral 3D case*)
```
producto exterior g··· deriva

In[∘]:=
$$\int_{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}}^{\frac{\sqrt[3]{k}}{\sqrt[3]{n}}} \frac{1}{8\left(-1+E^{\beta}\right)\pi r_k} 9\, n\, \phi \left(2-2k-\frac{6\, E^{\beta-\frac{(-1+k)^{2/3}\beta}{2 n^{2/3} r_k^2}} (-1+k)^{1/3} n^{2/3} r_k^2}{\beta}+\frac{3\, E^{\beta} n \sqrt{2\pi}\, \mathrm{Erf}\left[\frac{(-1+k)^{1/3}\sqrt{\beta}}{\sqrt{2}\, n^{1/3} r_k}\right] r_k^3}{\beta^{3/2}}\right) \alpha_v \, dr_k$$

Out[∘]=
$$\frac{1}{16\left(-1+E^{\beta}\right)\sqrt{2\pi}\,\beta^{3/2}} 9\, E^{\beta} n\, \phi \left(-4(-1+k)\,\mathrm{Erf}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]+4k\,\mathrm{Erf}\left[\frac{\left(\frac{-1+k}{k}\right)^{1/3}\sqrt{\beta}}{\sqrt{2}}\right]+\sqrt{\frac{2}{\pi}}\sqrt{\beta}\left(-2E^{-\beta/2}(-1+k)+2E^{-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta}(-1+k)^{1/3}k^{2/3}+(\beta-k\beta)\,\mathrm{ExpIntegralEi}\left[-\frac{\beta}{2}\right]+(-1+k)\beta\,\mathrm{ExpIntegralEi}\left[-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta\right]\right)\right)\alpha_v-\frac{1}{4\left(-1+E^{\beta}\right)\pi\beta}27(-1+k)^{1/3}n^{5/3}\phi\left(\frac{E^{\beta/2}(-1+k)^{2/3}\left(-2+E^{\beta/2}\beta\,\mathrm{Gamma}\left[0,\frac{\beta}{2}\right]\right)}{4n^{2/3}}+\frac{E^{\beta}\left(2E^{-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta}k^{2/3}-(-1+k)^{2/3}\beta\,\mathrm{Gamma}\left[0,\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta\right]\right)}{4n^{2/3}}\right)\alpha_v+\frac{9\,n\,\phi\left(-\mathrm{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]+\mathrm{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\alpha_v}{4\left(-1+E^{\beta}\right)\pi}-\frac{9\,k\,n\,\phi\left(-\mathrm{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]+\mathrm{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\alpha_v}{4\left(-1+E^{\beta}\right)\pi}$$

if condition

```
(*Integral for the k th photon 3D case*)
```

integral deriva

In[∘]:=
$$\int_{\frac{\sqrt[3]{k-1}}{\sqrt[3]{n}}}^{\frac{\sqrt[3]{k}}{\sqrt[3]{n}}} \frac{3}{4*\pi*r_k^{3}}*\alpha_v*\left(1-\left(\frac{1-E^{-\beta}*\frac{1}{2}}{1-E^{-\beta}}\right)\right)*\phi*3*r_k^{2}*n\,dr_k$$

Out[∘]=
$$\frac{9\,n\,\phi\left(-\mathrm{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]+\mathrm{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\alpha_v}{4\pi+4E^{\beta/2}\pi}$$ if condition

```
(*Summing the results up to  the k th photon*)
```

In[ ]:=

$$\frac{9\,n\,\phi\left(-\text{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]+\text{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\alpha_v}{4\,\pi+4\,E^{\beta/2}\,\pi}+$$

$$\frac{1}{16\left(-1+E^{\beta}\right)\sqrt{2\,\pi}\,\beta^{3/2}}\,9\,E^{\beta}\,n\,\phi\left(-4\,(-1+k)\,\text{Erf}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]+4\,k\,\text{Erf}\left[\frac{\left(\frac{-1+k}{k}\right)^{1/3}\sqrt{\beta}}{\sqrt{2}}\right]+\right.$$

función error función error

$$\sqrt{\frac{2}{\pi}}\,\sqrt{\beta}\left(-2\,E^{-\beta/2}\,(-1+k)+2\,E^{-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta}\,(-1+k)^{1/3}\,k^{2/3}+\right.$$

$$\left.\left.(\beta-k\,\beta)\,\text{ExpIntegralEi}\left[-\frac{\beta}{2}\right]+(-1+k)\,\beta\,\text{ExpIntegralEi}\left[-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta\right]\right)\right)\alpha_v-$$

Ei exponencial integral Ei exponencial integral

$$\frac{1}{4\left(-1+E^{\beta}\right)\pi\,\beta}\,27\,(-1+k)^{1/3}\,n^{5/3}\,\phi\left(\frac{E^{\beta/2}\,(-1+k)^{2/3}\left(-2+E^{\beta/2}\,\beta\,\text{Gamma}\left[0,\frac{\beta}{2}\right]\right)}{4\,n^{2/3}}+\right.$$

$$\left.\frac{E^{\beta}\left(2\,E^{-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta}\,k^{2/3}-(-1+k)^{2/3}\,\beta\,\text{Gamma}\left[0,\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta\right]\right)}{4\,n^{2/3}}\right)\alpha_v+$$

$$\frac{9\,n\,\phi\left(-\text{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]+\text{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\alpha_v}{4\left(-1+E^{\beta}\right)\pi}-\frac{9\,k\,n\,\phi\left(-\text{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]+\text{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\alpha_v}{4\left(-1+E^{\beta}\right)\pi}$$

Out[•]=
$$\frac{1}{16\left(-1+E^{\beta}\right)\sqrt{2\pi}\,\beta^{3/2}}\,9\,E^{\beta}\,n\,\phi\left(-4\,(-1+k)\,\mathrm{Erf}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]+4\,k\,\mathrm{Erf}\left[\frac{\left(\frac{-1+k}{k}\right)^{1/3}\sqrt{\beta}}{\sqrt{2}}\right]+\sqrt{\frac{2}{\pi}}\,\sqrt{\beta}\left(-2\,E^{-\beta/2}\,(-1+k)+2\,E^{-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta}\,(-1+k)^{1/3}\,k^{2/3}+(\beta-k\,\beta)\,\mathrm{ExpIntegralEi}\left[-\frac{\beta}{2}\right]+(-1+k)\,\beta\,\mathrm{ExpIntegralEi}\left[-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta\right]\right)\right)\alpha_v-\frac{1}{4\left(-1+E^{\beta}\right)\pi\,\beta}\,27\,(-1+k)^{1/3}\,n^{5/3}\,\phi\left(\frac{E^{\beta/2}\,(-1+k)^{2/3}\left(-2+E^{\beta/2}\,\beta\,\mathrm{Gamma}\left[0,\frac{\beta}{2}\right]\right)}{4\,n^{2/3}}+\frac{E^{\beta}\left(2\,E^{-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta}\,k^{2/3}-(-1+k)^{2/3}\,\beta\,\mathrm{Gamma}\left[0,\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta\right]\right)}{4\,n^{2/3}}\right)\alpha_v+\frac{9\,n\,\phi\left(-\mathrm{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]+\mathrm{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\alpha_v}{4\left(-1+E^{\beta}\right)\pi}-\frac{9\,k\,n\,\phi\left(-\mathrm{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]+\mathrm{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\alpha_v}{4\left(-1+E^{\beta}\right)\pi}+\frac{9\,n\,\phi\left(-\mathrm{Log}\left[\frac{(-1+k)^{1/3}}{n^{1/3}}\right]+\mathrm{Log}\left[\frac{k^{1/3}}{n^{1/3}}\right]\right)\alpha_v}{4\,\pi+4\,E^{\beta/2}\,\pi}$$

```
FullSimplify[%4] (*final result 3D case*)
```
simplifica completamente deriva

Out[•]=
$$\frac{1}{16\left(-1+E^{\beta}\right)\pi\,\beta^{3/2}}\,9\,n\,\phi\left(2\,E^{\beta}\sqrt{2\pi}+4\,E^{\beta/2}\,(-1+k)\,\sqrt{\beta}-4\,E^{\beta-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta}\,(-1+k)^{1/3}\,k^{2/3}\,\sqrt{\beta}+2\,E^{\beta}\,(-1+k)\,\sqrt{2\pi}\,\mathrm{Erfc}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]-2\,E^{\beta}\,k\,\sqrt{2\pi}\,\mathrm{Erfc}\left[\frac{\left(\frac{-1+k}{k}\right)^{1/3}\sqrt{\beta}}{\sqrt{2}}\right]+\frac{1}{3}\,\beta^{3/2}\left(-3\,E^{\beta}\,(-1+k)\left(3\,\mathrm{ExpIntegralE}\left[1,\frac{\beta}{2}\right]-3\,\mathrm{ExpIntegralE}\left[1,\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta\right]+\mathrm{ExpIntegralEi}\left[-\frac{\beta}{2}\right]-\mathrm{ExpIntegralEi}\left[-\frac{1}{2}\left(\frac{-1+k}{k}\right)^{2/3}\beta\right]\right)-4\,E^{\beta/2}\,(\mathrm{Log}[-1+k]-\mathrm{Log}[k])+4\,k\,(\mathrm{Log}[-1+k]-\mathrm{Log}[k])\right)\right)\alpha_v$$

```
(*Inner integral for the normalization constant 2D case*)
```
producto interior generalizado deriva

In[∘]:= $$\int_0^{r_k}\left(1-\left(\frac{1-E^{-\beta*r_i^2*\frac{1}{2*r_k^2}}}{1-E^{-\beta}}\right)\right)r_i\,dr_i$$

Out[∘]= $$\frac{\left(-2E^{\beta/2}+2E^{\beta}-\beta\right)r_k^2}{2\left(-1+E^{\beta}\right)\beta}$$

```
(*Outer integral for the normalization constant 2D case*)
```
producto exterior generalizado deriva

In[∘]:= $$\int_0^{2\pi}\frac{\left(-2E^{\beta/2}+2E^{\beta}-\beta\right)r_k^2}{2\left(-1+E^{\beta}\right)\beta}\,d\theta$$

$$\frac{\pi\left(-2E^{\beta/2}+2E^{\beta}-\beta\right)r_k^2}{\left(-1+E^{\beta}\right)\beta}$$ (* 2D Normalization *)

deriva

```
(*Inner integral for the normalization constant 3D case*)
```
producto interior generalizado deriva

In[∘]:= $$\int_0^{r_k}\left(1-\left(\frac{1-E^{-\beta*r_i^2*\frac{1}{2*r_k^2}}}{1-E^{-\beta}}\right)\right)r_i^2*\mathrm{Sin}[\phi]\,dr_i$$

seno

Out[∘]= $$\frac{\left(-2\sqrt{\beta}\left(3E^{\beta/2}+\beta\right)+3E^{\beta}\sqrt{2\pi}\,\mathrm{Erf}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]\right)\mathrm{Sin}[\phi]\,r_k^3}{6\left(-1+E^{\beta}\right)\beta^{3/2}}$$

```
(*Second inner integral for the normalization cosntant 3D case*)
```
deriva

In[∘]:= $$\int_0^{\pi}\frac{\left(-2\sqrt{\beta}\left(3E^{\beta/2}+\beta\right)+3E^{\beta}\sqrt{2\pi}\,\mathrm{Erf}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]\right)\mathrm{Sin}[\phi]\,r_k^3}{6\left(-1+E^{\beta}\right)\beta^{3/2}}\,d\phi$$

Out[∘]= $$\frac{\left(-2\sqrt{\beta}\left(3E^{\beta/2}+\beta\right)+3E^{\beta}\sqrt{2\pi}\,\mathrm{Erf}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]\right)r_k^3}{3\left(-1+E^{\beta}\right)\beta^{3/2}}$$

```
(*Outer integral for the normalization constant 3D case*)
```
producto exterior generalizado deriva

In[∘]:= $$\int_0^{2\pi}\frac{\left(-2\sqrt{\beta}\left(3E^{\beta/2}+\beta\right)+3E^{\beta}\sqrt{2\pi}\,\mathrm{Erf}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]\right)r_k^3}{3\left(-1+E^{\beta}\right)\beta^{3/2}}\,d\theta$$

$$\frac{2\pi\left(-2\sqrt{\beta}\left(3E^{\beta/2}+\beta\right)+3E^{\beta}\sqrt{2\pi}\,\mathrm{Erf}\left[\frac{\sqrt{\beta}}{\sqrt{2}}\right]\right)r_k^3}{3\left(-1+E^{\beta}\right)\beta^{3/2}}$$ (* 3D Normalization *)

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